

MACROSCOPIC MODELS FOR LARGE DYNAMIC SWARMS FOR IDENTIFICATION AND CONTROL: AN EULER ALIGNMENT SYSTEMS APPROACH*

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ABSTRACT. Euler alignment systems appear as hydrodynamic limits of interacting self-propelled particle systems and are particularly useful in the study of the macroscopic behavior of swarms. In this work, we provide an analysis of weak solutions to a class of Euler alignment systems in three spatial dimensions that corresponds to the limit of a large family of swarm dynamics that extends the (generalized) Cucker-Smale model. The models developed and analyzed incorporate the following novel features: bounded domain, singular interaction potential, confinement potential, and interaction with ODEs for control. We focus on Euler systems on smooth, bounded, and connected domains, possibly containing obstacles. We include confinement potential and self-propulsion, and singular alignment, attraction, and repulsion interaction kernels. We extend the interaction potentials from Yukawa to Bessel potentials of fractional order $s > 2$, and introduce an external influence term by leader-agents governed by controlled ODEs. We embed the problem into an abstract Euler system to conclude that infinitely many weak solutions exist. We show that we can construct solutions satisfying bounds on an energy quantity, and that the solutions satisfy a weak-strong uniqueness principle. We showcase the applicability of the Euler system under study, in the analysis of the behavior of large swarms. By solving a PDE-constrained optimization problem, we identify the interaction dynamics of a simulated swarm of agents from observations of their density evolution in 3D regions with obstacles. We close with several future directions and possible extensions including distributed control of swarms.

1. INTRODUCTION

In the last two decades, considerable interest has been shown in models of collective behavior across a variety of scientific fields, including traffic, ecology, robotics, and social networks. A particular area of this interest is in using insights gained from biological flocking and swarming to construct controls for large teams of UAVs [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]. As a result, a number of well-studied models of collective motion have been

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studied, including the Boids model, the Cucker-Smale model, and the Vicsek model [15, 16, 17]. These models assume behavioral rules at the individual level, such as velocity alignment, attraction, and repulsion, and can be seen as representatives of efforts by computer scientists, mathematicians/control theorists, and physicists, respectively, to describe the same phenomenon.

However, the microscopic analysis of large systems of agents presents many computational difficulties, and the transition to continuum limits has been proposed to alleviate some of them. In particular, there has been a substantial interest in macroscopic models that consider large number of interacting agents, approaching the mean-field limit. These models consist of kinetic and hydrodynamic (continuum) limits of interacting particle systems [18, 19, 20, 21, 22, 8]. that are described by PDEs defined on macroscopic quantities, such as the swarm's density.

In this work, we are interested in formal hydrodynamic limits of particle dynamics subject to velocity alignment, attraction, repulsion, self-propulsion, and containment forces in bounded, smooth regions. Similar particle systems arise in design of controls for autonomous vehicles or UAVs using virtual forces [6, 23] or virtual potentials [11, 12, 13, 14]. Alignment and cohesion provide useful team-like properties to the dynamics. It is desirable to construct virtual forces between UAVs which prevent collision between them, hence the repulsion is included. Collision between a UAV and obstacles or confinement to a desired region is enforced by a boundary on their dynamics and encouraged by a confinement potential. The boundary need not be physical for the UAVs: it can be a virtual boundary designed to enforce safety constraints. The self-propulsion is added to counteract the dissipation introduced by the velocity alignment term.

We first study a problem on bounded region $D \subset \mathbb{R}^3$ with the Yukawa potential providing alignment, attraction, and repulsion. Then, we incorporate leaders into our model, converting these PDEs into a coupled ODE-PDE system. Bessel potentials of fractional order $s > 2$ are also discussed. Finally, we showcase the application of our analysis by developing a numerical solver for a PDE-constrained optimization problem that reconstructs the interaction dynamics of a large swarm of agents interacting within a workspace cluttered with obstacles, via observations of their density, i.e., their probability distribution in three spatial dimensions, plus time.

The identification of coordination laws of swarms from partially observed data (microscopic or macroscopic) is a motivating key problem studied in recent previous works by the authors and others [7, 8, 24, 25, 26]. The distributed robust control of swarms is another motivating key problem studied in previous works by the authors and others [11, 12, 13, 14, 6, 9, 10, 23]. For both problems understanding the nature of solutions of the corresponding systems of dynamical equations and their uniqueness or non-uniqueness is fundamental. Even more important problems for which substantial additional work is needed are the discovery of the simplest possible coordination

laws and the robustness of such coordination laws to perturbations and noise.

The emphasis of this paper is on developing and incorporating additional and challenging features in the problem formulation and model development corresponding to practical applications, and then developing appropriate extensions/modifications of previously developed mathematical analysis techniques, or additional elements, for the analysis of these new models.

1.1. Related Work. Most work on Euler alignment systems with singular interactions focus on the Riesz potential [27, 28] or a similar mildly singular kernel [29]. [28, 29] focus on the 1D model in the whole space $\mathbb{R}$ with singular alignment force only, and obtain global existence of classical solutions. Previous work also exists on hydrodynamic limits of particle dynamics with fairly regular nonlocal alignment and attraction-repulsion kernels over the torus [30].

However, in this work, we show that weak solutions exist to systems with singular alignment, attraction, and repulsion kernels and a bounded domain possibly containing obstacles. This work is motivated by the need to develop a robust theory for a particular application to system identification problems [8] in three spatial dimensions [24]. Therefore, we are not interested in the problem including pressure and temperature, as in [30, 22]. Our analysis is based on extensions of the work of [30, 31] based on results from [32, 33].

Regarding the application of our analysis, in our previous work, we discussed the tradeoffs associated to particle and density-based approaches to discovering swarm interaction laws [7], and explored this problem in two spatial dimensions, plus time [8]. Accurate and efficient solvers for the hydrodynamic Cucker-Smale model were developed which employed the Kurganov-Tadmor finite volume method [34] to compute the hyperbolic part of the PDEs, and spectral solvers which computed the nonlocal part. Our work was influenced by [25], where the authors followed a similar approach modeling the interactions with respect to a fractional differential system of equations. More recently, the same problem is studied from the perspective of physics-informed neural networks [26].

1.2. Contribution. In this work, we show existence of infinitely many weak solutions of Euler alignment systems on a bounded, smooth domain with singular alignment, attraction, and repulsion kernels and a bounded domain possibly containing obstacles. We then show that these solutions satisfy an energy inequality, and are hence bounded-energy weak solutions. We use this energy inequality to prove a weak-strong uniqueness principle.

To the best of our knowledge, this is one of the first works on this type of 3D Euler alignment system in bounded domains with obstacles. Many existing works on similar models are posed on the free space or the torus [35, 36, 37]. In addition, we maintain a variety of interactions in our Euler alignment system, local and non-local, including alignment, attraction, repulsion, obstacles, and self-propulsion, while, most existing work on the

Euler alignment system and similar systems focus on a subset of these [27, 28, 29, 35, 36, 37]. Finally, the (non-fractional) Bessel-type interaction which we select for our alignment, attraction, and repulsion terms has not been studied before, as is the case for the introduction of leaders.

Moreover, we further showcase the applicability of the Euler alignment system under review, in the problem of discovering the interaction (or coordination) dynamics of a large swarm of agents interacting within a workspace cluttered with obstacles, via observations of their density, i.e., their probability distribution in three spatial dimensions, plus time [24]. We generate 3D particle trajectories using a version of the Boids model [15] in a cubical region of space with obstacles. From these trajectories, we extract time-evolving spatial probability distribution estimates that are then used as continuous observations of the density of the swarm. To discover the interaction (or coordination) laws, which we parametrize with a real vector, we pose a PDE-constrained optimization problem to minimize the squared Hellinger distance between the Boid histogram and the hydrodynamic mean-field distribution. We solve this problem numerically and approximately using the Newton-conjugate gradient method. To the authors' knowledge, this is the first instance of a macroscopic multi-agent system identification problem for flocking (i.e. identification of distributed coordination laws) being solved in three spatial dimensions in the presence of obstacles.

1.3. Paper Organization. In Section 2, we formally describe the transition from a microscopic Cucker-Smale-type particle/agent model to a macroscopic Euler alignment system, to be used in our consequent analysis. In Section 3, we state and prove our results. First, we show via convex integration that distributional solutions to the Euler alignment system exist (Section 3.1). Then, we show that these solutions satisfy some energy bounds, and so are bounded-energy weak solutions (Section 3.2). Finally, we use the energy inequality to show that the solutions satisfy a weak-strong uniqueness principle, i.e. that a Lipschitz solution, should it exist, is unique in the class of bounded-energy weak solutions to the Euler alignment system (Section 3.3). In Section 4, we first expand the results we have to include leader-agents which interact with the agent-fluid described by the Euler alignment system (Section 4.1), and show that the conclusions of Section 3 hold for this system, as well. We then describe an Euler alignment system subject to a broader class of non-local fractional Bessel-type interactions, and show the conclusions of our previous results in Section 3 hold for this class of models (Section 4.2). Finally, in Section 5, we showcase the applicability of the Euler system under study in the identification problem of large swarm dynamics. Section 6 concludes the paper.

1.4. Notation. We use the ellipsis punctuation ‘...’ to indicate line continuation for long equations or expressions. We also use $C \in \mathbb{R}^+$ to denote several generic constants, and denote any dependence of this constant on other parameters (constants, functions, or sets) $\alpha_1, \dots, \alpha_n$ by: $C(\alpha_1, \dots, \alpha_n)$.

We use the notation $C_w([t_0, t_f]; X)$ to mean functions $f : [t_0, t_f] \rightarrow X$, X a reflexive Banach space, s.t. $\forall \{t_n\}_{n=1}^\infty \subset [t_0, t_f] : t_n \rightarrow t$ in $[t_0, t_f]$ f is s.t.:

$$(f(t_n) - f(t), \psi)_{X, X^*} \rightarrow 0 \text{ for all } \psi \in X^*,$$

where $(\cdot, \cdot)_{X, X^*}$ is the pairing of an element of X with an element of its dual space X^* . A similar notation is used in [30, 31, 32], and a slightly different one in [33, 38]. We use the following notations for the inner product and outer products for real vectors $v, w \in \mathbb{R}^3$:

$$v^\top w \equiv \langle v, w \rangle_{\mathbb{R}^3}, vw^\top \equiv v \otimes w,$$

where $v^\top$ is the transposition of v . We denote $\text{cl } S$ to be the closure of the set $S \subset \mathbb{R}^3$, and we use $l : \mathbf{L}(S) \rightarrow \mathbb{R}_0^+$ to denote the standard Lebesgue measure, $\mathbf{L}(S)$ being the σ -algebra of Lebesgue-measurable sets contained in S . If S is bounded and smooth(-enough), $\nu : \partial S \rightarrow \partial B(0, 1)$ is the outward pointing normal on the boundary ∂S . We use $\mathbb{I}_A : \mathbb{R}^3 \rightarrow \{0, 1\}$ to denote the characteristic (indicator) function of a set $A \subset \mathbf{L}(\mathbb{R}^3)$.

2. HYDRODYNAMIC ALIGNMENT SYSTEMS

In this section, we introduce a particle model of a multi-agent system and its formal kinetic and hydrodynamic limits. For certain very regular interactions, it can be shown fairly readily that the kinetic and hydrodynamic (under a suitable closure assumption) behaviors indeed correspond to limits related to the empirical measure of the particle dynamics. Singular kernels present significant obstacles to demonstrating such a correspondence [27, 39].

Consider a sequence of particles (boids) $\{(x_i, v_i)\}_{i=1}^N(t) \subset (\text{cl } D \times \mathbb{R}^3)$, $t \in [t_0, t_f]$, $D \subset \mathbb{R}^3$ a bounded, open, connected, smooth region. The particles are associated to probability space $\{\Omega, \mathcal{B}(\Omega), \mathbb{P}\}$ and state space $\{(\text{cl } D)^N \times \mathbb{R}^{3N}, \mathcal{B}((\text{cl } D)^N \times \mathbb{R}^{3N})\}$. The equations of motion for the particles are for each $\omega \in \Omega$:

$$(2.1) \quad \begin{cases} \frac{d}{dt} x_i(t) = v_i(t) \\ \frac{d}{dt} v_i(t) = \sum_{\alpha \in A} \Phi_\alpha(x_i(t), v_i(t), \{(x_i, v_i)\}_{i=1}^N(t)) \\ x_i(t) \in \text{cl } D \quad \forall t \in [t_0, t_f] \end{cases}$$

with i.i.d. initial conditions $(x_i, v_i)(t_0) \sim \mathbb{M}_{xv} \in \mathcal{P}(D \times \mathbb{R}^3)$. $A := \{a, c, r, p, o\}$ are character subscripts corresponding to alignment, cohesion, repulsion,

self-propulsion, and obstacle avoidance forces. These forces $\Phi_\alpha(\cdots)$ are:

$$\begin{aligned}
(2.2) \quad \Phi_a(x_i, v_i, \{(x_i, v_i)\}_{i=1}^N) &:= \frac{1}{N-1} \sum_{j=1, j \neq i}^N G(x_i, x_j; k_a, \lambda_a)(v_j - v_i), \\
\Phi_c(x_i, v_i, \{(x_i, v_i)\}_{i=1}^N) &:= \frac{1}{N-1} \sum_{j=1, j \neq i}^N \nabla_1 G(x_i, x_j; k_c, \lambda_c), \\
\Phi_r(x_i, v_i, \{(x_i, v_i)\}_{i=1}^N) &:= \frac{1}{N-1} \sum_{j=1, j \neq i}^N \nabla_1 G(x_i, x_j; k_r, \lambda_r), \\
\Phi_p(x_i, v_i, \{(x_i, v_i)\}_{i=1}^N) &= \Phi_p(v_i) := \kappa_p v_i (1 - P(\|v_i\|_{\mathbb{R}^3})) \\
\Phi_o(x_i, v_i, \{(x_i, v_i)\}_{i=1}^N) &= \Phi_o(x_i) := \nabla_1 U(x_i),
\end{aligned}$$

where $\nabla_1(\cdot)$ is the gradient with respect to the first (or only) argument, and $k_\alpha, \lambda_\alpha, \kappa_p > 0$ are real constants. The class of kernels $G : D \times \mathbb{R}^3 \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ represent interaction functions between each pair of particles. As will be explained in Lemma 2.1, following our previous analysis in [8], we are interested in symmetric interaction functions that constitute Green's functions associated with certain elliptic partial differential operators (Assumption 1). We take:

$$(2.3) \quad P \in C_b^2(\mathbb{R}; \mathbb{R}), P'(x) \geq 0 \text{ s.t. } \Phi_p \in \text{Lip}(\mathbb{R}^3; \mathbb{R}^3), \text{ and } U \in C_b^2(D; \mathbb{R}).$$

To satisfy the spatial constraint, we impose specular reflections on ∂D :

$$(2.4) \quad \lim_{t_{i,j}^{*+} \downarrow t_{i,j}^*} v_j(t_{i,j}^{*+}) = (\mathbf{I}_3 - 2\nu(x_j(t_{i,j}^*))\nu^\top(x_j(t_{i,j}^*)))v_j(t_{i,j}^*),$$

where $t_{i,j}^*$ is the time of the i -th collision of the j -th agent with the boundary ∂D .

Let $\mu(t) \in \mathcal{P}(D \times \mathbb{R}^3)$, $\mu(t_0) = \mathbb{M}_{xv}$ be the single-particle position-velocity probability distribution [40] of the generalized Cucker-Smale ensemble at time t . Assume it has density ρ_{xv} w.r.t. the Lebesgue measure. Similarly to [40], the formal kinetic (Vlasov) limit of the given particle dynamics is:

$$(2.5) \quad \begin{cases} \partial_t \rho_{xv} + \nabla_x \cdot [v \rho_{xv}] + \nabla_v \cdot [\mathcal{F}[\rho_{xv}](t, \cdot) \rho_{xv}] = 0 \\ \mathcal{F}[\rho_{xv}](t, x, v) := \int_{D \times \mathbb{R}^3} G(x, y; k_a, \lambda_a)(w - v) \rho_{xv}(t, y, w) dy dw + \dots \\ \int_D \nabla_x G(x, y; k_r, \lambda_r) \rho_{xv}(t, y, v) dy + \dots \\ \int_D \nabla_x G(x, y; k_c, \lambda_c) \rho_{xv}(t, y, v) dy + \dots \\ \nabla_x U(x) + \kappa_p v (1 - P(\|v\|_{\mathbb{R}^3})) \end{cases}$$

with boundary conditions on $[t_0, t_f] \times \partial D$ [40, 41]:

$$\rho_{xv}(t, x, v) = \rho_{xv}(t, x, (\mathbf{I}_3 - 2\nu(x)\nu^\top(x))v),$$

and initial condition $\rho_{xv}(t_0, \cdot) = \rho_0^{xv}$, which is the density associated to $\mathbb{M}_{xv}$. Now, we define the spatial probability density and momentum density:

$$\rho : [t_0, t_f] \times D \rightarrow \mathbb{R}, \mathbf{j} : [t_0, t_f] \times D \rightarrow \mathbb{R}^3,$$

respectively, as:

$$(2.6) \quad \begin{aligned} \rho(t, \cdot) &:= \int_{\mathbb{R}^3} \rho_{xv}(t, \cdot) dv \\ \mathbf{j}(t, \cdot) &:= \int_{\mathbb{R}^3} v \rho_{xv}(t, \cdot) dv =: \rho(t, \cdot) \mathbf{u}(t, \cdot). \end{aligned}$$

Assume $\rho > 0$ in D , and consider $(\rho(t, \cdot), \mathbf{j}(t, \cdot))^\top$ to be extended by zero outside D when we use them as functions over the whole $\mathbb{R}^3$. It can be shown after repeated formal application of integration by parts and assumption of the mono-kinetic closure *ansatz* [3]:

$$(2.7) \quad \rho_{xv}(t, x, v) = \rho(t, x) \delta(v - \mathbf{u}(t, x))$$

that we obtain the compressible Euler equations over $(t, x) \in [t_0, t_f] \times D$:

$$(2.8) \quad \begin{cases} \partial_t \rho + \nabla_x \cdot \mathbf{j} = 0 & \text{in } (t_0, t_f] \times D \\ \partial_t \mathbf{j} + \nabla_x \cdot [\rho^{-1} \mathbf{j} \mathbf{j}^\top] = \mathbf{S}[\rho, \mathbf{j}] & \text{in } (t_0, t_f] \times D \\ \mathbf{S}[\rho, \mathbf{j}] = \rho \mathcal{L}_a \mathbf{j} - \mathbf{j} \mathcal{L}_a \rho - \rho \nabla_x [\mathcal{L}_c \rho + \mathcal{L}_r \rho + U] + \kappa_p \mathbf{j} (1 - P(\rho^{-1} \|\mathbf{j}\|_{\mathbb{R}^3})), \end{cases}$$

subject to the non-penetrative boundary conditions [41] on ∂D and initial conditions:

$$(2.9) \quad \mathbf{j}^\top \nu = 0 \quad \forall (t, x) \in [t_0, t_f] \times \partial D; \quad (\rho(t_0, \cdot), \mathbf{j}(t_0, \cdot))^\top = (\rho_0, \mathbf{j}_0)^\top \text{ in } D.$$

This is the Euler alignment system. $\mathcal{L}_\alpha : L^2(D; \mathbb{R}) \rightarrow L^2(D; \mathbb{R})$ denotes a linear operator of form

$$\mathcal{L}_\alpha \phi \circ (t, \cdot) := \int_{\mathbb{R}^3} G(\cdot, y; k_\alpha; \lambda_\alpha) \mathbb{I}_D(y) \phi(t, y) dy.$$

If we take

$$G(\cdot; k_a, \lambda_a), G(\cdot; k_c, \lambda_c), G(\cdot; k_r, \lambda_r)$$

to be fundamental solutions for elliptic partial differential operators $\Lambda_a, \Lambda_c, \Lambda_r$, respectively, we can construct an augmented system of PDEs for state vector field $(\rho, \mathbf{j})$ with alignment potentials $\pi_\rho[\rho], \pi_{\mathbf{j}}[\mathbf{j}]$ and cohesion-repulsion

potentials $\mathbf{V}[\rho] := (v_c[\rho], v_r[\rho])^\top$:

$$(2.10) \quad \begin{cases} \partial_t \rho + \nabla_x \cdot \mathbf{j} = 0 & \text{in } (t_0, t_f] \times D; \\ \partial_t \mathbf{j} + \nabla_x \cdot [\rho^{-1} \mathbf{j} \mathbf{j}^\top] = \mathbf{S}[\rho, \mathbf{j}] & \text{in } (t_0, t_f] \times D; \\ \Lambda_a \pi_\rho[\rho](t, \cdot) = \mathbb{I}_D \rho(t, \cdot) & \text{in } \mathbb{R}^3; \\ \Lambda_a \pi_{\mathbf{j}}[\mathbf{j}](t, \cdot) = \mathbb{I}_D \mathbf{j}(t, \cdot) & \text{in } \mathbb{R}^3; \\ \Lambda_c v_c[\rho](t, \cdot) = \mathbb{I}_D \rho(t, \cdot) & \text{in } \mathbb{R}^3; \\ \Lambda_r v_r[\rho](t, \cdot) = \mathbb{I}_D \rho(t, \cdot) & \text{in } \mathbb{R}^3; \\ \mathbf{S}[\rho, \mathbf{j}] := \rho \pi_{\mathbf{j}}[\mathbf{j}] - \mathbf{j} \pi_\rho[\rho] - \rho \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U] + \kappa_p \mathbf{j} (1 - P(\rho^{-1} \|\mathbf{j}\|_{\mathbb{R}^3})); \\ \mathbf{j}^\top \nu = 0 & \text{in } [t_0, t_f] \times \partial D; \\ (\rho(t_0, \cdot), \mathbf{j}(t_0, \cdot))^\top = (\rho_0, \mathbf{j}_0)^\top & \text{in } D, \end{cases}$$

where $\mathbf{1}_n$ is the n -vector with ones as its entries. We pick the parametrized family of linear elliptic operators with constant coefficients:

$$(2.11) \quad \mathcal{L}_\alpha^{-1} = \Lambda_\alpha = \frac{1}{4\pi k_\alpha} (\lambda_\alpha^2 I - \nabla_x^2),$$

with $k_a, k_r > 0, k_c < 0, \lambda_a, \lambda_r, \lambda_c \neq 0$. The dynamical system (2.10) is one of the first developed for such dynamic swarms with the additional features mentioned and obstacles. It is a macroscopic model of the swarm incorporating specific functional form of the various potentials. More general forms of potentials have been introduced in associated control problems in previous works by the authors [11, 12, 13, 14], including robustness of the resulting dynamics to the actual functional form of the potentials. That is, the resulting dynamics under control are very similar as long as the various potential functions have appropriate qualitative properties [11, 12, 13, 14]. The resulting dynamic optimal control under such potentials is randomized.

We present a very simple lemma which will aid us in showing existence of weak solutions to this augmented system later on.

Lemma 2.1. *Consider the elliptic problem:*

$$\Lambda_\alpha w[\phi] = \mathbb{I}_D \phi \text{ in } \mathbb{R}^3$$

with $k_\alpha, \lambda_\alpha \neq 0$. If $\phi \in L^\infty(D; \mathbb{R})$, then $w \in C_b^1(D; \mathbb{R})$. If:

$$\phi \in C^{0,\alpha}(D; \mathbb{R}), \alpha \in (0, 1]$$

then $\phi \in C^{2,\alpha}(D; \mathbb{R})$.

Proof. (Sketch.) Notice the fundamental solution is the Yukawa potential:

$$G(\|x - y\|_{\mathbb{R}^3}; k_\alpha, \lambda_\alpha) = k_\alpha \frac{e^{-\lambda_\alpha \|x - y\|_{\mathbb{R}^3}}}{\|x - y\|_{\mathbb{R}^3}} = 4\pi k_\alpha e^{-\lambda_\alpha \|x - y\|_{\mathbb{R}^3}} G_N(\|x - y\|_{\mathbb{R}^3})$$

where G_N is the Newtonian potential. The Newtonian potential clearly is also an upper bound to the Yukawa potential. A solution to the above is:

$$(2.12) \quad w[\phi](\cdot) = (G(\cdot; k_\alpha, \lambda_\alpha) * \mathbb{I}_D v)(\cdot) = \int_{\mathbb{R}^3} G(\cdot - y; k_\alpha, \lambda_\alpha) \mathbb{I}_D(y) \phi(y) dy.$$

We can use the same approaches to obtain regularity for the Newtonian potential [42] on the Yukawa potential and the results of Lemma 2.1 follow. $\square$

Remark 2.2. We can prove this Lemma via results on Bessel potential spaces and conclude $w[v] \in C^{2,\alpha}(D; \mathbb{R})$ assuming $v \in H^2(D; \mathbb{R})$ and then using a Sobolev embedding to obtain Hölder regularity. See Theorem 4.3 for details.

Assumption 1. Throughout the rest of the manuscript, we will assume that the potentials $\pi_\rho, \pi_{\mathbf{j}}, v_r, v_c$ are computed via the given convolution integral representation (2.12) for each time $t \in [t_0, t_f]$.

3. EXISTENCE AND WEAK-STRONG UNIQUENESS

In this section, we describe weak solutions to the system of PDEs (2.10). The solutions $(\rho, \mathbf{j})$ are of type:

$$(3.1) \quad \begin{aligned} \rho &\in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}), \rho > 0, \\ \mathbf{j} &\in C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)) \cap L^\infty((t_0, t_f) \times D; \mathbb{R}^3) \end{aligned}$$

which satisfy (2.10) in the following sense:

$$(3.2) \quad \int_D \rho(t, \cdot) \phi(t, \cdot) \, dx \Big|_{t_0}^{t_f} = \int_{t_0}^{t_f} \int_D \rho \partial_t \phi + \mathbf{j}^\top \nabla_x \phi \, dx \, dt;$$

$$(3.3) \quad \begin{aligned} \int_D \mathbf{j}^\top(t, \cdot) \psi(t, \cdot) \, dx \Big|_{t_0}^{t_f} = \dots \\ \int_{t_0}^{t_f} \int_D \mathbf{j}^\top \partial_t \psi + \langle \rho^{-1} \mathbf{j} \mathbf{j}^\top, \nabla_x \psi \rangle_{\mathbb{R}^{3 \times 3}} + \psi^\top \mathbf{S}[\rho, \mathbf{j}] \, dx \, dt; \end{aligned}$$

$$(3.4) \quad \rho(t_0, \cdot) = \rho_0, \mathbf{j}(t_0, \cdot) = \mathbf{j}_0$$

for arbitrary $(\phi, \psi) \in \mathcal{D}([t_0, t_f] \times D; \mathbb{R} \times \mathbb{R}^3)$. Throughout, assume:

$$\rho_0 \in H^2(D; \mathbb{R}) \cap \text{Lip}(\text{cl } D; \mathbb{R}), \rho_0 > 0,$$

$$\mathbf{j}_0 \in \text{Lip}(\text{cl } D; \mathbb{R}^3),$$

and consider $(\rho(t, \cdot), \mathbf{j}(t, \cdot))^\top$ to be extended by zero outside D if we must use them as functions over the whole $\mathbb{R}^3$.

3.1. Existence of “Wild” Weak Solutions.

Theorem 3.1. *Consider the system of PDEs (2.10). Suppose $\rho_0, \mathbf{j}_0, U, P$ are (as assumed earlier):*

$$P \in C_b^2(\mathbb{R}; \mathbb{R}), P'(x) \geq 0 \text{ s.t. } \Phi_p \in \text{Lip}(\mathbb{R}^3; \mathbb{R}^3), \text{ and } U \in C_b^2(D; \mathbb{R}),$$

$$\rho_0 \in H^2(D; \mathbb{R}) \cap \text{Lip}(\text{cl } D; \mathbb{R}), \rho_0 > 0,$$

$$\mathbf{j}_0 \in \text{Lip}(\text{cl } D; \mathbb{R}^3).$$

Then, (2.10) admits infinitely many weak solutions of the type:

$$(3.5) \quad \begin{aligned} \rho &\in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap Lip([t_0, t_f] \times cl D; \mathbb{R}), \rho > 0, \\ \mathbf{j} &\in C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)) \cap L^\infty((t_0, t_f) \times D; \mathbb{R}^3). \end{aligned}$$

Proof. To prove this result, we embed the problem into an abstract functional *incompressible* Euler system [31, 32, 33] and extend and apply methods and results from there. First, define the binary operation $(\cdot \odot \cdot) : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{S}_0^3$, the (constant) zero-trace symmetric matrices:

$$v \odot w := vw^\top - \frac{1}{3}v^\top w \mathbf{I}_3.$$

We seek to transform our problem into: Find

$$\mathbf{v} \in \left(C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)) \cap L^\infty((t_0, t_f) \times D; \mathbb{R}^3) \right) =: X$$

s.t. (for some $\mathbf{h}, r, \mathbf{M}$ of a class soon-to-be-specified):

$$(3.6) \quad \partial_t \mathbf{v} + \nabla_x \cdot \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) \odot (\mathbf{v} + \mathbf{h}[\mathbf{v}])}{r[\mathbf{v}]} + \mathbf{M}[\mathbf{v}] \right] = 0$$

in the sense of $\mathcal{D}'([t_0, t_f] \times D; \mathbb{R}^3)$;

$$(3.7) \quad \nabla_x \cdot \mathbf{v}(t, \cdot) = 0$$

(after an extension by zero to $\mathbb{R}^3$) in the sense of $\mathcal{D}'([t_0, t_f] \times \mathbb{R}^3; \mathbb{R}^3)$;

$$(3.8) \quad \mathbf{v}^\top(t, \cdot)\nu = 0 \text{ on } \partial D \ \forall t \in [t_0, t_f];$$

$$(3.9) \quad \mathbf{v}(t_0, \cdot) = \mathbf{v}_0;$$

subject to the constraint:

$$(3.10) \quad \frac{1}{2} \frac{\|\mathbf{v} + \mathbf{h}[\mathbf{v}]\|_{\mathbb{R}^3}^2}{r[\mathbf{v}]}(\cdot) = e[\mathbf{v}](\cdot) \text{ a.e. in } [t_0, t_f] \times D$$

for some e whose class is likewise soon-to-be-specified. It is assumed for open set $Q \subseteq (t_0, t_f) \times D$ s.t. $l(Q) = l((t_0, t_f) \times D)$ that:

$$\mathbf{h} : X \rightarrow C_b(Q; \mathbb{R}^3),$$

$$r : X \rightarrow C_b(Q; \mathbb{R}), r > 0,$$

$$e : X \rightarrow C_b(Q; \mathbb{R}), e \geq 0, \text{ and}$$

$$\mathbf{M} : X \rightarrow C_b(Q; \mathbb{S}_0^3)$$

are Q -continuous, which is to say:

Definition 3.2. An operator

$$b : X \rightarrow C_b(Q; \mathcal{V}),$$

$\mathcal{V}$ a finite dimensional vector space, is **Q-continuous**, if b is s.t.:

- (1) b maps sets bounded in $L^\infty((t_0, t_f) \times D; \mathbb{R}^3)$ to sets bounded in $C_b(Q; \mathcal{V})$;

- (2) For $\{\mathbf{w}_m\}_{m=1}^\infty$, $\mathbf{w}_m \in X$, and $\mathbf{w}_m \rightarrow \mathbf{w} \in X$ weakly in $C_w([t_0, t_f]; L^2(D; \mathbb{R}^3))$ and weakly-* in $L^\infty((t_0, t_f) \times D; \mathbb{R}^3)$ implies

$$b[\mathbf{w}_m] \rightarrow b[\mathbf{w}] \text{ uniformly in } C_b(Q; \mathcal{V});$$

- (3) If $\mathbf{v}(t, \cdot) = \mathbf{w}(t, \cdot)$ for $t_0 \leq t \leq \tau \leq t_f$, then:

$$b[\mathbf{v}] = b[\mathbf{w}] \text{ in } ((0, \tau] \times D) \cap Q.$$

We introduce a set of subsolutions to the abstract system (3.6- 3.9), $X_0 \subset X$, where for $\mathbf{F} \in L^\infty((t_0, t_f) \times D; \mathbb{S}_0^3)$, $\mathbf{v} \in X$, $\mathbf{h}, \mathbf{F}, r, e, \mathbf{M}$ are Q -continuous and s.t for $t_0 < t < t_f$:

$$(3.11) \quad \frac{3}{2} \lambda_{max} \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) \odot (\mathbf{v} + \mathbf{h}[\mathbf{v}])}{r[\mathbf{v}]} + \mathbf{M}[\mathbf{v}] - \mathbf{F} \right] (t, \cdot) - e[\mathbf{v}](t, \cdot) < 0,$$

$$X_0 := \left\{ \mathbf{v} : (3.8 - 3.11) \text{ satisfied, } \partial_t \mathbf{v} + \nabla_x \cdot \mathbf{F} = 0 \text{ in } \mathcal{D}'([t_0, t_f] \times D; \mathbb{R}^3) \right\}.$$

A pair $(\mathbf{v}, \mathbf{F})$ which satisfy these constraints is termed an *admissible subsolution-flux pair*.

Similarly to [30, 31], we have the following theorem:

Theorem 3.3. *Suppose operators $\mathbf{h}, r, e$, and $\mathbf{M}$ are Q -continuous and subject to the conditions (3.11). Suppose that $(r[\mathbf{v}])^{-1}$ is Q -continuous. Further, suppose X_0 is non-empty and bounded in $L^\infty((t_0, t_f) \times D; \mathbb{R}^3)$. Then, the abstract system (3.6- 3.10) admits infinitely many weak solutions.*

Proof. Define

$$e[\mathbf{v}](t, \cdot) := \sigma(t) - \frac{3}{2} \Sigma[\mathbf{v}](t, \cdot)$$

for some Q -continuous operator Σ , and some $\sigma : [t_0, t_f] \rightarrow \mathbb{R}$ to be determined s.t. $e[\mathbf{v}] \geq 0$, and

$$\eta[\mathbf{v}, \mathbf{F}](t, \cdot) := \frac{3}{2} \lambda_{max} \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) (\mathbf{v} + \mathbf{h}[\mathbf{v}])^\top}{r[\mathbf{v}]} + \mathbf{M}[\mathbf{v}] - \mathbf{F} \right] (t, \cdot).$$

To begin the proof, we present a Lemma, similar to Lemma 3.2 in [33]:

Lemma 3.4. *Suppose $\{(\mathbf{v}_m, \mathbf{F}_m)\}_{m=1}^\infty$ are a sequence of admissible subsolution-flux pairs. Then, $\exists$ a unique $(\mathbf{v}^*, \mathbf{F}^*)$ s.t.*

(3.12)

$$\mathbf{v}_m \rightarrow \mathbf{v}^* \text{ in } C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)), \text{ and weakly-* in } L^\infty((t_0, t_f) \times D; \mathbb{R}^3);$$

$$\mathbf{F}_m \rightarrow \mathbf{F}^* \text{ weakly-* in } L^\infty((t_0, t_f) \times D; \mathbb{S}_0^3),$$

which is an admissible subsolution-flux pair to the abstract Euler problem up to the relaxation on $t_0 < t < t_f$:

(3.13)

$$\frac{3}{2} \lambda_{max} \left[\frac{(\mathbf{v}^* + \mathbf{h}[\mathbf{v}^*]) \odot (\mathbf{v}^* + \mathbf{h}[\mathbf{v}^*])}{r[\mathbf{v}^*]} + \mathbf{M}[\mathbf{v}^*] - \mathbf{F}^* \right] (t, \cdot) - e[\mathbf{v}^*](t, \cdot) \leq 0.$$

Further, if for the functional $I : L^2([t_0, t_f] \times D; \mathbb{R}^3) \times (0, (t_f - t_0)/2) \rightarrow \mathbb{R}_0^-$:

$$(3.14) \quad I[\mathbf{v}](t) := \inf_{\tau \in [t_0+t, t_f-t]} \int_D \frac{1}{2} \frac{\|\mathbf{v} + \mathbf{h}[\mathbf{v}]\|_{\mathbb{R}^3}^2}{r[\mathbf{v}]}(\tau, \cdot) - e[\mathbf{v}](\tau, \cdot) \, dx$$

$I[\mathbf{v}^*](t) \equiv 0$, then $\mathbf{v}^*$ solves the abstract Euler system (3.6- 3.10).

Proof. (sketch) The claim follows with minor modification to the argument for Lemma 3.2. in [33]. Simply replace the construction of the operator $e[\cdot]$ and the functional $I[\cdot]$ therein with the ones defined here and the claim follows. $\square$

Next, we give the *Oscillatory Lemma* of [31, 33].

Lemma 3.5. (*Oscillatory Lemma*) Let $\mathbf{v} \in X_0$ and $(\mathbf{v}, \mathbf{F})$ be an admissible subsolution-flux pair, and suppose $t \in (t_0, t_f)$. Then, $\exists \{\mathbf{w}_m\}_{m=1}^\infty \subset \mathcal{D}((t, t_f) \times D; \mathbb{R}^3), \{\Psi_m\}_{m=1}^\infty \subset \mathcal{D}((t, t_f) \times D; \mathbb{S}_0^3)$ s.t.

$(\mathbf{v} + \mathbf{w}_m, \mathbf{F} + \Psi_m)$ are an admissible subsolution-flux pair

$\mathbf{w}_m \rightarrow 0$ in $C_w([t_0, t_f]; L^2(D; \mathbb{R}^3))$

there exists $C(E) > 0$ s.t. $e[\mathbf{v}](t) \leq E$ and

$\liminf_{m \rightarrow \infty} I[\mathbf{v} + \mathbf{w}_m](t) \geq I[\mathbf{v}](t) + C(E)I^2[\mathbf{v}](t).$

Proof. Let $\tilde{X} := C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)) \cap C((t_0, t_f) \times D; \mathbb{R}^3)$, and:

$f \in C_b((t_0, t_f) \times D; \mathbb{R}); \mathbf{F} \in C_b((t_0, t_f) \times D; \mathbb{S}_0^3),$

$$(3.15) \quad \begin{aligned} \Xi(\mathbf{F}, f) := \Big\{ & \mathbf{v} \in \tilde{X} : \text{there exists } \mathbf{\Upsilon} \in C_b((t_0, t_f) \times D; \mathbb{S}_0^3) \text{ s.t. } \dots \\ & \partial_t \mathbf{v} + \nabla_x \cdot \mathbf{\Upsilon} = 0 \text{ in } \mathcal{D}'([t_0, t_f] \times D; \mathbb{R}^3), \dots \\ & \mathbf{v}^\top \nu = 0 \text{ on } \partial D \, \forall t \in [t_0, t_f], \dots \\ & \forall \tau > t_0 \inf_{(t,x) \in (\tau, t_f) \times D} f(t, x) - \epsilon[\mathbf{v}, \mathbf{\Upsilon}, \mathbf{F}](t, x) > 0, \dots \\ & \mathbf{v}(t_0, \cdot) = \mathbf{v}_0 \Big\}, \end{aligned}$$

where $\epsilon : \tilde{X} \times C_b((t_0, t_f) \times D; \mathbb{S}_0^3) \times C_b((t_0, t_f) \times D; \mathbb{S}_0^3) \rightarrow C_b(Q; \mathbb{R})$ is

$$(3.16) \quad \epsilon[\mathbf{v}, \mathbf{\Upsilon}, \mathbf{F}] := \frac{3}{2} \lambda_{\max} \left(\frac{1}{r[\mathbf{v}]} \mathbf{v} \mathbf{v}^\top + \mathbf{F} - \mathbf{\Upsilon} \right).$$

To continue the proof of the *Oscillatory Lemma*, we present a Lemma similar to the one presented in [33], which was originally proven in [32]:

Lemma 3.6. Let $\mathcal{A} := (t_1, t_2) \times D \subseteq Q$ be an open subset, and $\mathbf{F}, f$ of the classes previously specified, $f > 0$ in $(t_0, t_f) \times D$. Suppose $\mathbf{v} \in \Xi(\mathbf{F}, f)$ with flux $\mathbf{\Upsilon}$. Then, there exists $\Lambda > 0$ depending only on f and sequences

$$\begin{aligned}
(3.17) \quad & \{\mathbf{w}_m\}_{m=1}^\infty \subset \mathcal{D}(\mathcal{A}; \mathbb{R}^3), \{\Psi_m\}_{m=1}^\infty \subset \mathcal{D}(\mathcal{A}; \mathbb{S}_0^3) \text{ s.t.} \\
& \mathbf{v} + \mathbf{w}_m \in \Xi(\Gamma, f) \text{ with flux } \Upsilon + \Psi_m \\
& \mathbf{w}_m \rightarrow 0 \text{ in } C_w([t_0, t_f]; L^2(D; \mathbb{R}^3)) \\
& \liminf_{m \rightarrow \infty} \inf_{t \in (t_1, t_2)} \int_D \frac{1}{2} \frac{1}{r[\mathbf{v}]} \|\mathbf{v} + \mathbf{w}_m\|_{\mathbb{R}^3}^2 dx \geq \dots \\
& \inf_{t \in (t_1, t_2)} \left[\int_D \frac{1}{2} \frac{1}{r[\mathbf{v}]} \|\mathbf{v}\|_{\mathbb{R}^3}^2 dx + \Lambda \left(\int_D f - \frac{1}{2} \frac{1}{r[\mathbf{v}]} \|\mathbf{v}\|_{\mathbb{R}^3}^2 dx \right)^2 \right]
\end{aligned}$$

The rest of the proof of Lemma 3.5 follows directly from the arguments given in [33]. $\square$

Finally, we use the results on Baire categories first used by [38] and applied in [31, 33], and others:

Lemma 3.7. *Let (X, d) be a complete metric space and suppose a function $I : X \rightarrow \mathbb{R}_0^-$ is a Baire class-1 function. Suppose $X_0 \subseteq X$ is dense in X and non-empty, s.t. $\forall \beta < 0 \exists \alpha(\beta) > 0, \forall x \in X_0$ with:*

$$I(x) < \beta < 0 \exists \{x_m\}_{m=1}^\infty \in X_0, x_m \rightarrow x \text{ in } (X, d)$$

and

$$\liminf_{m \rightarrow \infty} I(x_m) \geq I(x) + \alpha(\beta)$$

then $\exists S \subseteq X_0$ s.t. $I(x) = 0$ on S .

We now follow the rest of the result of [33] and the proof of Theorem 3.3 is concluded. $\square$

Now, it remains to construct the operators $r, \mathbf{h}, e, \mathbf{M}$ as they pertain to our problem. Take $Q := (t_0, t_f) \times D$. We take the Helmholtz decomposition for $\mathbf{j}$:

$$\mathbf{j} = \mathbf{v} + \nabla_x \Phi[\mathbf{v}]$$

similarly to the procedure of [43]. To satisfy the boundary conditions (BCs) on $\mathbf{j}$ as well as the continuity equation for ρ , we take $\Phi[\mathbf{v}]$ as the solution of the Neumann problem:

$$\begin{aligned}
(3.18) \quad & \nabla_x^2 \Phi[\mathbf{v}] = -\partial_t \rho[\mathbf{v}] \text{ in } D \\
& (\nabla_x^\top \Phi[\mathbf{v}])\nu = 0 \text{ on } \partial D.
\end{aligned}$$

Using the Helmholtz decomposition, we have expressed the momentum in such a way that we can select $\rho[\mathbf{v}]$ as we need it. Select an arbitrary

$$\rho \in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}), \rho > 0, \rho(t_0, \cdot) = \rho_0$$

s.t. $\int_D \rho(t, \cdot) dx = 1$. Now, rather obviously using the dominated convergence theorem on one hand we have:

$$\frac{d}{dt} \int_D \rho(t, \cdot) dx = \int_D \partial_t \rho(t, \cdot) dx,$$

and on the other hand we know $\int_D \rho(t, \cdot) dx = 1$. So, we conclude

$$\int_D \partial_t \rho(t, \cdot) dx = 0.$$

Since the R.H.S. of (3.18) and the associated boundary condition have the same integral over their respective domains for each $t \in [t_0, t_f]$, the Neumann problem (3.18) has a unique $H^1(D; \mathbb{R})$ solution up to an additive constant for each $t \in [t_0, t_f]$ via Lax-Milgram [42]. It is sufficient to select any one of these solutions. Additionally, we have that:

$$\Phi[\mathbf{v}](t, \cdot) \in H^4(D; \mathbb{R}) \subset C^{2,1/2}(D; \mathbb{R})$$

using elliptic regularity and Sobolev embedding [44, 45]. $\nabla_x \Phi[\mathbf{v}]$ is thus unique. From thrice time-differentiability of $\rho[\mathbf{v}]$:

$$\Phi[\mathbf{v}] \in C^2([t_0, t_f]; C^{2,1/2}(D; \mathbb{R})).$$

The Helmholtz decomposition of $\mathbf{j}_0$ is then:

$$\mathbf{j}_0 = \mathbf{v}_0 + \nabla_x \Phi[\mathbf{v}](t_0, \cdot)$$

so we can obtain $\mathbf{v}_0$ from $\rho_0, \mathbf{j}_0$. The assumption on $\rho[\mathbf{v}]$ guarantees that via the integral representation outlined in Lemma 2.1, the repulsion and attraction potentials are s.t. (via a Sobolev embedding):

$$(3.19) \quad v_r[\rho[\mathbf{v}]](t, \cdot), v_c[\rho[\mathbf{v}]](t, \cdot) \in C^{2,1/2}(D; \mathbb{R}) \text{ for each } t \in [t_0, t_f],$$

so the gradients of these potentials exist in the classical sense. Note that also:

$$\pi_\rho[\rho[\mathbf{v}]](t, \cdot) \in C^{2,1/2}(D; \mathbb{R}), \pi_{\mathbf{j}}[\mathbf{v} + \nabla_x \Phi[\mathbf{v}]] \in C^{2,1/2}(D; \mathbb{R}^3) \text{ for each } t \in [t_0, t_f]$$

via Lemma 2.1 by the assumptions on $\rho[\mathbf{v}](t, \cdot)$ and $\mathbf{v} \in X$. We re-write the momentum equation using the Helmholtz decomposition:

$$(3.20) \quad \begin{aligned} \partial_t \mathbf{v} + \nabla_x \cdot \left[\frac{(\mathbf{v} + \nabla_x \Phi[\mathbf{v}])(\mathbf{v} + \nabla_x \Phi[\mathbf{v}])^\top}{\rho[\mathbf{v}]} + \mathbf{I} \partial_t \Phi[\mathbf{v}] \right] &= \mathbf{S}[\cdots] \\ \mathbf{S}[\rho[\mathbf{v}], \mathbf{v} + \nabla_x \Phi[\mathbf{v}]] &= \rho[\mathbf{v}] \pi_{\mathbf{j}}[\mathbf{v} + \nabla_x \Phi[\mathbf{v}]] - \dots \\ (\mathbf{v} + \nabla_x \Phi[\mathbf{v}]) \pi_\rho[\rho[\mathbf{v}]] &- \rho[\mathbf{v}] \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho[\mathbf{v}]] + U] + \dots \\ \kappa_p(\mathbf{v} + \nabla_x \Phi[\mathbf{v}]) &(1 - P(\rho^{-1}[\mathbf{v}] \|\mathbf{v} + \nabla_x \Phi[\mathbf{v}]\|_{\mathbb{R}^3})). \end{aligned}$$

Let:

$$\mathbf{h}[\mathbf{v}] = \nabla_x \Phi[\mathbf{v}],$$

which is clearly Q -continuous (it is constant in $\mathbf{v}$). From Theorem 3.3, we want $\mathbf{v}$ over Q to be subject to the constraint:

$$\frac{1}{2} \frac{\|\mathbf{v} + \mathbf{h}[\mathbf{v}]\|_{\mathbb{R}^3}^2}{\rho[\mathbf{v}]} = e[\mathbf{v}](t, \cdot) \equiv \sigma[\mathbf{v}](t) - \frac{3}{2} \Sigma[\mathbf{v}](t, \cdot) \text{ a.e. on } Q.$$

Take $\Sigma[\mathbf{v}] = \partial_t \Phi[\mathbf{v}]$, and assume $\sigma(\cdot)$ is continuous and bounded. We will specify σ further later on. From the R.H.S., $e[\mathbf{v}]$ is clearly Q -continuous. We modify the momentum equation again using this “kinetic energy” :

$$\begin{aligned}
 (3.21) \quad & \partial_t \mathbf{v} + \nabla_x \cdot \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) \odot (\mathbf{v} + \mathbf{h}[\mathbf{v}])}{\rho[\mathbf{v}]} \right] = \mathbf{S}[\dots] \\
 & \mathbf{S}[\rho[\mathbf{v}], \mathbf{v} + \mathbf{h}[\mathbf{v}]] = \rho[\mathbf{v}] \pi_j \left[\mathbf{v} + \mathbf{h}[\mathbf{v}] \right] - \dots \\
 & (\mathbf{v} + \mathbf{h}[\mathbf{v}]) \pi_\rho[\rho[\mathbf{v}]] - \rho[\mathbf{v}] \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho[\mathbf{v}]] + U] + \dots \\
 & \kappa_p(\mathbf{v} + \mathbf{h}[\mathbf{v}]) (1 - P(\sqrt{2} \rho^{-1/2}[\mathbf{v}] e^{1/2}[\mathbf{v}])).
 \end{aligned}$$

Consider now the second-order elliptic homogeneous Dirichlet problem :

$$\begin{aligned}
 (3.22) \quad & -\nabla_x \cdot \left[\nabla_x \mathbf{w}[\mathbf{v}] + \nabla_x^\top \mathbf{w}[\mathbf{v}] - \frac{2}{3} \mathbf{I} \operatorname{tr} \nabla_x \mathbf{w}[\mathbf{v}] \right] = \mathbf{S}[\rho[\mathbf{v}], \mathbf{j}[\mathbf{v}]] \text{ in } D; \\
 & \mathbf{w}[\mathbf{v}] = 0 \text{ on } \partial D
 \end{aligned}$$

which is linear in $\mathbf{w}$. Let

$$(3.23) \quad \mathbf{M}[\mathbf{v}] = \nabla_x \mathbf{w}[\mathbf{v}] + \nabla_x^\top \mathbf{w}[\mathbf{v}] - \frac{2}{3} \mathbf{I} \operatorname{tr} \nabla_x \mathbf{w}[\mathbf{v}],$$

which is trace-free symmetric. Define the bilinear form $\mathbf{B} : H_0^1(D; \mathbb{R}^3) \times H_0^1(D; \mathbb{R}^3) \rightarrow \mathbb{R}$:

$$\mathbf{B}[\chi, \phi] := \langle \nabla_x \chi + \nabla_x^\top \chi - \frac{2}{3} \mathbf{I} \operatorname{tr} \nabla_x \chi, \nabla_x \phi \rangle_{L^2(D; \mathbb{R}^{3 \times 3})}.$$

Over matrices of form $\xi \eta^\top, \xi, \eta \in \mathbb{R}^3$, we can show using the elementary symmetric polynomials of the matrix characteristic polynomial:

$$\langle \xi \eta^\top + \eta \xi^\top - \frac{2}{3} \operatorname{tr} \xi \eta^\top \mathbf{I}, \xi \eta^\top \rangle_{\mathbb{R}^{3 \times 3}} = 2 \|\xi \eta^\top\|_{\mathbb{R}^{3 \times 3}}^2 - \frac{2}{3} [\operatorname{tr} \xi \eta^\top]^2 = \frac{4}{3} \|\xi \eta^\top\|_{\mathbb{R}^{3 \times 3}}^2$$

since these matrices are rank-1. Hence, the (constant-coefficient) elliptic operator in the given problem satisfies the Legendre-Hadamard condition, and then by Garding's theorem [46] the associated bilinear form is coercive. Now, we apply the Lax-Milgram theorem to conclude that the PDE:

$$\mathbf{B}[\chi, \phi] = (\mathbf{S}[\rho[\mathbf{v}], \mathbf{v} + \mathbf{h}[\mathbf{v}]], \phi)_{H^{-1}(D; \mathbb{R}^3), H_0^1(D; \mathbb{R}^3)}$$

with $\mathbf{S}[\rho[\mathbf{v}](t, \cdot), \mathbf{j}[\mathbf{v}](t, \cdot)] \in L^2(D; \mathbb{R}^3)$ has a unique $H_0^1(D; \mathbb{R}^3)$ solution χ for each $t \in [t_0, t_f]$. Now, we apply this result to $\mathbf{w}[\mathbf{v}]$, and conclude that $\mathbf{w}[\mathbf{v}] \in H_0^1(D; \mathbb{R}^3)$ exists and is unique. By boundary regularity [46] of the constant-coefficient problem, $\mathbf{w}[\mathbf{v}](t, \cdot) \in H_0^2(D; \mathbb{R}^3)$. $\mathbf{w}$ is also a continuous linear map from $L^p(D; \mathbb{R}^3)$ into $W^{2,p}(D; \mathbb{R}^3)$. See Lemma 7.1., Corr. 7.2. of [33] for details. Since $\mathbf{S}[\rho[\mathbf{v}], \mathbf{v} + \mathbf{h}[\mathbf{v}]]$ is also $L^\infty((t_0, t_f) \times D; \mathbb{R}^3)$, we conclude that $\mathbf{w}[\mathbf{v}] \in L^\infty(t_0, t_f; W^{2,p}(D; \mathbb{R}^3))$ for $p \in (1, \infty)$. It can be verified that the operator $\mathbf{S}$ reformulated as we have done using Φ, e preserves weak continuity in $L^2(D; \mathbb{R}^3)$ and weak* continuity in $L^\infty(D; \mathbb{R}^3)$ for each $t \in$

$[t_0, t_f]$. Since $\mathbf{v} \in C_w([t_0, t_f]; L^2(D; \mathbb{R}^3))$, $\mathbf{w} \in C_w([t_0, t_f]; L^2(D; \mathbb{R}^3))$ by continuity, so $\mathbf{w}$ is s.t. $\forall \psi \in L^2(D; \mathbb{R}^3)$:

$$\begin{aligned} \infty &> \langle \mathbf{w}(t, \cdot) \Big|_{t_0}^{t_f}, \psi \rangle_{L^2(D; \mathbb{R}^3)} = \int_D \int_{t_0}^{t_f} \partial_t (\mathbf{w}^\top(t, \cdot) \psi) dt dx = \dots \\ &\quad \int_{t_0}^{t_f} \int_D \partial_t \mathbf{w}^\top(t, \cdot) \psi dx dt, \end{aligned}$$

so $\partial_t \mathbf{w} \in L^2(t_0, t_f; L^2(\tilde{D}; \mathbb{R}^3))$. We have the embeddings [47]:

$$W^{2,4}(D; \mathbb{R}^3) \subset\subset C_b^1(D; \mathbb{R}^3) \subset L^2(D; \mathbb{R}^3).$$

By the Aubin-Lions-Simon theorem (Cor. 4 of [48]),

$$\mathbf{w}[X] \subset\subset C([t_0, t_f]; C_b^1(D; \mathbb{R}^3)),$$

so $\mathbf{M} \in C_b(Q; \mathbb{S}_0^3) \forall \mathbf{v} \in X$. The last property for Q -continuity of this operator is easily verified. Hence, we finally have:

$$(3.24) \quad \partial_t \mathbf{v} + \nabla_x \cdot \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) \odot (\mathbf{v} + \mathbf{h}[\mathbf{v}])}{\rho[\mathbf{v}]} + \mathbf{M}[\mathbf{v}] \right] = 0.$$

$r[\mathbf{v}] := \rho[\mathbf{v}]$ is obviously Q -continuous, as is $(r[\mathbf{v}])^{-1}$, and, as we have noted, so are e , $\mathbf{h}$, and $\mathbf{M}$. Let $\mathbf{F} \equiv 0$, i.e. $\mathbf{v}(t, \cdot) = \mathbf{v}_0$ a.e.. So, the set X_0 is non-empty by our choices. We can select a function $\sigma \in C_b([t_0, t_f]; \mathbb{R})$ s.t.:

$$(3.25) \quad \sup_{(t,x) \in Q, t > \tau} \frac{3}{2} \left[\lambda_{max} \left[\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) \odot (\mathbf{v} + \mathbf{h}[\mathbf{v}])}{r[\mathbf{v}]} + \mathbf{M}[\mathbf{v}] \right] + \partial_t \Phi[\mathbf{v}] \right] < \sigma[\mathbf{v}](t)$$

since the L.H.S. is bounded. Now, since we have selected σ , we employ the inequality similar to the one given in Lemma 3.2 of [38]:

$$\frac{1}{2} \frac{\|\mathbf{v} + \mathbf{h}[\mathbf{v}]\|_{\mathbb{R}^3}^2}{r} \leq \frac{3}{2} \lambda_{max} \left(\frac{(\mathbf{v} + \mathbf{h}[\mathbf{v}]) (\mathbf{v} + \mathbf{h}[\mathbf{v}])^\top}{r} - \mathbf{U} \right) \text{ if } \mathbf{U} \in \mathbb{S}_0^3$$

to conclude X_0 is bounded in $L^\infty((t_0, t_f) \times D; \mathbb{R}^3)$. We invoke Theorem 3.3 and the claim of Theorem 3.1 is proven. $\square$

3.2. Bounded-Energy Weak Solutions. We now consider a more specific variety of weak solution to this problem: the bounded-energy weak solution. We construct a quantity which we show is bounded along solutions to the the system (2.10). First, we show some formal computations to motivate the construction of the energy inequality similarly to [30, 31]. Recall $\mathbf{u} = \rho^{-1} \mathbf{j}$. We begin:

$$\partial_t [\rho \mathbf{u}^\top \mathbf{u}] = [\partial_t (\rho \mathbf{u})]^\top \mathbf{u} + (\rho \mathbf{u})^\top \partial_t \mathbf{u}.$$

Multiplying the momentum equation in (2.10) by $\mathbf{u}$ gives:

$$\begin{aligned} \mathbf{u}^\top \partial_t (\rho \mathbf{u}) + \mathbf{u}^\top \nabla_x \cdot [\rho \mathbf{u} \mathbf{u}^\top] &= \rho \mathbf{u}^\top \pi_{\mathbf{j}}[\rho \mathbf{u}] - \rho \mathbf{u}^\top \mathbf{u} \pi_\rho[\rho] - \dots \\ \rho \mathbf{u}^\top \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U] &+ \kappa_p \rho \mathbf{u}^\top \mathbf{u} [1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})]. \end{aligned}$$

Substituting using the relation for $\partial_t[\rho \mathbf{u}^\top \mathbf{u}]$ yields:

$$\begin{aligned} \partial_t[\rho \mathbf{u}^\top \mathbf{u}] - (\rho \mathbf{u})^\top \partial_t \mathbf{u} + \mathbf{u}^\top \nabla_x \cdot [\rho \mathbf{u} \mathbf{u}^\top] &= \dots \\ \rho \mathbf{u}^\top \pi_j[\rho \mathbf{u}] - \rho \mathbf{u}^\top \mathbf{u} \pi_\rho[\rho] - \rho \mathbf{u}^\top \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U] + \kappa_p \rho \mathbf{u}^\top \mathbf{u} [1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})] & \end{aligned}$$

From the continuity and momentum equations, with $\rho > 0$, formally:

$$\partial_t \mathbf{u} + (\nabla_x^\top \mathbf{u}) \mathbf{u} = \pi_j[\rho \mathbf{u}] - \mathbf{u} \pi_\rho[\rho] - \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U] + \kappa_p \mathbf{u} (1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})).$$

Substituting these terms into the previous computation and applying integration by parts formally, we obtain:

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \int_D \rho \mathbf{u}^\top \mathbf{u} \, dx &= \int_D \rho \mathbf{u}^\top \pi_j[\rho \mathbf{u}] - \rho \mathbf{u}^\top \mathbf{u} \pi_\rho[\rho] \, dx + \dots \\ &\quad \int_D -\rho \mathbf{u}^\top \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U] + \kappa_p \rho \mathbf{u}^\top \mathbf{u} [1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})] \, dx. \end{aligned}$$

Notice that formally:

$$\int_D \rho \mathbf{u}^\top \nabla_x v_r[\rho] \, dx = - \int_D \nabla_x \cdot [\rho \mathbf{u}] v_r[\rho] \, dx = \frac{1}{2} \frac{d}{dt} \int_D \rho v_r[\rho] \, dx$$

and similarly for v_c . So, we now obtain the (strict) energy balance after re-expressing the equations in $(\rho, \mathbf{j})$:

$$\begin{aligned} (3.26) \quad \frac{1}{2} \frac{d}{dt} \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} + \rho v_r[\rho] + \rho v_c[\rho] \, dx &= \dots \\ \int_D \mathbf{j}^\top \pi_j[\mathbf{j}] - \rho^{-1} \mathbf{j}^\top \mathbf{j} \pi_\rho[\rho] + \mathbf{j}^\top \nabla_x U + \kappa_p \rho^{-1} \mathbf{j}^\top \mathbf{j} [1 - P(\|\rho^{-1} \mathbf{j}\|_{\mathbb{R}^3})] \, dx & \end{aligned}$$

We have now constructed a suitable energy quantity to study. We now relax the strictness of what we obtained to accommodate weak solutions to the system (2.10). Similarly to [30, 31, 49], we have the definition:

Definition 3.8. A pair $(\rho, \mathbf{j})$ is a *bounded-energy weak solution* of the system (2.10) iff:

- (1) $(\rho, \mathbf{j})$ are of type:

$$\begin{aligned} \rho &\in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}), \rho > 0; \\ \mathbf{j} &\in X; \end{aligned}$$

- (2) The solutions satisfy (3.2-3.4);

- (3) The solution additionally satisfies the relaxed energy balance:

$$\begin{aligned} \frac{1}{2} \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j}(t, \cdot) + \rho v_r[\rho](t, \cdot) + \rho v_c[\rho](t, \cdot) \, dx \Big|_{t_0}^\tau &\leq \dots \\ \int_{t_0}^\tau \int_D \mathbf{j}^\top \pi_j[\mathbf{j}] - \rho^{-1} \mathbf{j}^\top \mathbf{j} \pi_\rho[\rho] \, dx \, dt + \dots & \\ \int_{t_0}^\tau \int_D \mathbf{j}^\top \nabla_x U + \kappa_p \rho^{-1} \mathbf{j}^\top \mathbf{j} (1 - P(\|\rho^{-1} \mathbf{j}\|_{\mathbb{R}^3})) \, dx \, dt & \end{aligned}$$

a.e. on $[t_0, t_f]$.

Theorem 3.9. *Consider the problem (2.10). Suppose $\rho_0, \mathbf{j}_0, U, P$ are (as assumed earlier):*

$$P \in C_b^2(\mathbb{R}; \mathbb{R}), P'(x) \geq 0 \text{ s.t. } \Phi_p \in Lip(\mathbb{R}^3; \mathbb{R}^3), \text{ and } U \in C_b^2(D; \mathbb{R}),$$

$$\rho_0 \in H^2(D; \mathbb{R}) \cap Lip(\text{cl } D; \mathbb{R}), \rho_0 > 0,$$

$$\mathbf{j}_0 \in Lip(\text{cl } D; \mathbb{R}^3).$$

Then, (2.10) admits infinitely many bounded-energy weak solutions.

Proof. As in Theorem 3.1, we can select densities ρ to be as regular as we desire. So, as before, select $\rho \in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap Lip([t_0, t_f] \times \text{cl } D; \mathbb{R})$. Then, construct potential $\Phi[\mathbf{v}]$, and we conclude that there are infinitely many fields $\mathbf{v}$ which satisfy the abstract Euler problem in the weak sense. From the construction of the energy operator $e[\mathbf{v}]$, we know that a.e. on $[t_0, t_f] \times D$:

$$(3.27) \quad \frac{1}{2} \frac{\|\mathbf{j}\|_{\mathbb{R}^3}^2}{\rho}(t, x) = e[\mathbf{v}](t, x) \equiv \sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}](t, x).$$

So, we know that:

$$\begin{aligned} & \frac{1}{2} \frac{d}{dt} \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} + \rho v_r[\rho] + \rho v_c[\rho] \, dx = \dots \\ & \frac{1}{2} \frac{d}{dt} \int_D 2(\sigma(t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) + \rho v_r[\rho] + \rho v_c[\rho] \, dx. \end{aligned}$$

Considering the R.H.S. of the strict energy balance (3.26), the last term can be bounded easily as:

$$\begin{aligned} \kappa_p \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} (1 - P(\|\rho^{-1} \mathbf{j}\|_{\mathbb{R}^3})) \, dx &\leq C(\kappa_p, P) \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} \, dx = \dots \\ & 2C(\kappa_p, P) \int_D (\sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) \, dx, \end{aligned}$$

as $(1 - P)$ is uniformly bounded. For the second-to-last term, we note that there exist $\tilde{\rho}, \bar{\rho}$ s.t.

$$0 < \tilde{\rho} < \rho < \bar{\rho} < \infty \text{ everywhere in } [t_0, t_f] \times D,$$

and so:

$$(\bar{\rho})^{-1/2} \|f\|_{L^2(D; \mathbb{R}^3)} \leq \|\rho^{-1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)} \leq (\tilde{\rho})^{-1/2} \|f\|_{L^2(D; \mathbb{R}^3)}$$

for $f \in L^2(D; \mathbb{R}^3)$. $\|\rho^{-1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)}$ additionally satisfies the properties of norms - it is obviously non-negative, $\|\rho^{1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)} = 0$ iff $f \equiv_{L^2(D; \mathbb{R}^3)} 0$, and satisfies:

$$\|\rho^{-1/2}(t, \cdot) (f + g)\|_{L^2(D; \mathbb{R}^3)} \leq \|\rho^{-1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)} + \|\rho^{-1/2}(t, \cdot) g\|_{L^2(D; \mathbb{R}^3)}$$

$$\|C \rho^{-1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)} \leq |C| \|\rho^{-1/2}(t, \cdot) f\|_{L^2(D; \mathbb{R}^3)}.$$

So, $\|\rho^{-1/2}(t, *) (\cdot)\|_{L^2(D; \mathbb{R}^3)}$ is equivalent to the standard norm on $L^2(D; \mathbb{R}^3)$. Accordingly, using the Cauchy-Schwarz inequality, the equivalence we obtained, and Young's inequality:

$$\begin{aligned} \int_D \mathbf{j}^\top \nabla_x U \, dx &\leq C(U) \|\mathbf{j}\|_{L^2(D; \mathbb{R}^3)} \leq C(U) \bar{\rho}^{1/2} \|\rho^{-1/2}(t, \cdot) \mathbf{j}(t, \cdot)\|_{L^2(D; \mathbb{R}^3)} \leq \dots \\ \frac{1}{2} C(U, \bar{\rho})^2 + \frac{1}{2} \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} \, dx &= \frac{1}{2} C(U, \bar{\rho})^2 + \int_D (\sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) \, dx. \end{aligned}$$

On the alignment terms, we can apply the triangle inequality, Young's convolution inequality, our equivalence of norms, and the boundedness of $\pi_\rho[\rho]$ to obtain:

$$\begin{aligned} \int_D \mathbf{j}^\top \pi_j[\mathbf{j}] - \rho^{-1} \mathbf{j}^\top \mathbf{j} \pi_\rho[\rho] \, dx &\leq C(k_a, \lambda_a, \tilde{\rho}, \bar{\rho}) \|\rho^{-1/2}(t, \cdot) \mathbf{j}(t, \cdot)\|_{L^2(D; \mathbb{R}^3)}^2 = \dots \\ 2C(k_a, \lambda_a, \tilde{\rho}, \bar{\rho}) \int_D (\sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) \, dx \end{aligned}$$

so we arrive at:

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \int_D 2(\sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) + \rho v_r[\rho] + \rho v_c[\rho] \, dx &\leq \dots \\ C_1(\kappa_p, P, U, \tilde{\rho}, \bar{\rho}, k_a, \lambda_a) (1 + \int_D (\sigma[\mathbf{v}](t) - \frac{3}{2} \partial_t \Phi[\mathbf{v}]) \, dx). \end{aligned}$$

We move the potential terms to the R.H.S., which gives:

$$\begin{aligned} \frac{d}{dt} \int_D \sigma[\mathbf{v}](t) \, dx &= l(D) \frac{d}{dt} \sigma[\mathbf{v}](t) \leq \dots \\ C_1(\dots) \left(1 + \int_D \sigma[\mathbf{v}](t) + \frac{3}{2} \partial_t \Phi[\mathbf{v}] \, dx \right) &+ \dots \\ \left| \frac{1}{2} \frac{d}{dt} \int_D \rho v_r[\rho] + \rho v_c[\rho] - 3 \partial_t \Phi[\mathbf{v}] \, dx \right| &\leq C_2(D, C_1(\dots)) (1 + \sigma[\mathbf{v}](t)) \end{aligned}$$

by time-differentiability and spatial boundedness in D of the relevant functions. These spatial bounds depend only on the choice of $\tilde{\rho}, \bar{\rho}$ which are the relevant bounds on ρ . Thus, selection of σ which does not increase 'too quickly' to satisfy a bound on the energy is feasible. It remains to choose such a σ . We do so similarly to [30] Let σ_0, b be positive, real constants sufficiently large so that the differential inequality we obtained is satisfied by σ of the form:

$$\sigma[\mathbf{v}](t) = \sigma_0 + \exp(-b(t - t_0)) > \frac{3}{2} \|\partial_t \Phi[\mathbf{v}]\|_{C_b([t_0, t_f] \times D; \mathbb{R})},$$

so that $e[\mathbf{v}](t) > 0$, and

(3.28)

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \int_D \rho^{-1} \mathbf{j}^\top \mathbf{j} + \rho v_r[\rho] + \rho v_c[\rho] \, dx &\leq \dots \\ \int_D \mathbf{j}^\top \pi_j[\mathbf{j}] - \rho^{-1} \mathbf{j}^\top \mathbf{j} \pi_\rho[\rho] + \mathbf{j}^\top \nabla_x U + \kappa_p \rho^{-1} \mathbf{j}^\top \mathbf{j} [1 - P(\|\rho^{-1} \mathbf{j}\|_{\mathbb{R}^3})] \, dx. \end{aligned}$$

It remains to examine if the L.H.S. is discontinuous at t_0 . To conclude it is not, we use a slightly modified version of Theorem 6.1 of [31] given as Lemma 6.2 in [33]. The proof is identical to the one given in [31].

Theorem 3.10. *Let $\mathbf{M}$ as defined in (3.23) satisfy the hypotheses of Theorem 3.3, and suppose*

$$l(\{x \in D : (t, x) \in Q\}) = l(D).$$

Then, $\exists$ a dense subset $\mathcal{R} \subset (t_0, t_f)$ s.t. $\forall t \in \mathcal{R} \exists \mathbf{v} \in X$ s.t.:

$$\mathbf{v} \in C(((t_0, t) \cup (t, t_f)) \times D; \mathbb{R}^3) \cap C_w([t_0, t_f]; L^2(D; \mathbb{R}^3));$$

$$\mathbf{v}(t_0, \cdot) = \mathbf{v}_0;$$

$$\exists \mathbf{F} \in C(((t_0, t) \cup (t, t_f)) \times D; \mathbb{S}_0^3) \text{ s.t. } (\mathbf{v}, \mathbf{F})$$

is an admissible subsolution-flux pair;

$$(e[\mathbf{v}] - \eta[\mathbf{v}, \mathbf{F}])(t, x) > 0 \text{ in } ((t_0, t) \cup (t, t_f)) \times D;$$

$$\frac{1}{2} \frac{\|\mathbf{v} + \mathbf{h}[\mathbf{v}]\|_{\mathbb{R}^3}^2}{\rho[\mathbf{v}]} = e[\mathbf{v}] \text{ a.e. in } ((t_0, t) \cup (t, t_f)) \times D.$$

Thus, the claim of Theorem 3.9 follows with $Q = (t_0, t_f) \times D$. $\square$

3.3. Weak-Strong Uniqueness. We have established that there are solutions which satisfy energy bounds. Now, we show that if there exists a strong solution within the larger class of bounded-energy weak solutions, it is unique.

Our approach is slightly different than the usual ones taken for weak-strong uniqueness. Generally, for compressible Euler systems similar to 2.10 with (infinitely many) distributional solutions $(\rho, \mathbf{j})$, we would show that any strong solution $(r, \mathbf{i})$ would be unique in the class of weak solutions, assuming it exists.

However, we are not even interested in the case where ρ is distributional - our definition of weak solution in (3.1 - 3.4) is already restricted to only admit strong solutions for ρ . So, our weak-strong uniqueness principle is only for $\mathbf{j}$, reflected in our choice of relative energy.

Theorem 3.11. *Consider the system (2.10), and suppose $\rho_0, \mathbf{j}_0, U, P$ are (as assumed earlier):*

$$P \in C_b^2(\mathbb{R}; \mathbb{R}), P'(x) \geq 0 \text{ s.t. } \Phi_p \in Lip(\mathbb{R}^3; \mathbb{R}^3), \text{ and } U \in C_b^2(D; \mathbb{R}),$$

$$\rho_0 \in H^2(D; \mathbb{R}) \cap Lip(cl D; \mathbb{R}), \rho_0 > 0,$$

$$\mathbf{j}_0 \in Lip(cl D; \mathbb{R}^3).$$

Select a density of type:

$$\rho \in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap Lip([t_0, t_f] \times cl D; \mathbb{R}), \rho > 0, \rho(t_0, \cdot) = \rho_0,$$

and apply the results of Theorem 3.9 to obtain infinitely many bounded-energy weak solutions $(\rho, \mathbf{j})$. Then, assuming there exists a momentum field

$$(r, \mathbf{i}) \in Lip([t_0, t_f] \times cl D; \mathbb{R} \times \mathbb{R}^3), (r, \mathbf{i})(t_0, \cdot) = (\rho_0, \mathbf{j}_0)$$

which is a strong solution to (2.10), every $(\rho, \mathbf{j})$ corresponding to a bounded-energy weak solution is s.t.

$$(\rho, \mathbf{j}) = (r, \mathbf{i}) \text{ a.e. in } [t_0, t_f] \times D.$$

Proof. To prove this result, we define a relative energy quantity. Let $\mathbf{u} = \rho^{-1}\mathbf{j}$, $\tilde{\mathbf{u}} := r^{-1}\mathbf{i}$. Note that $\tilde{\mathbf{u}} \in \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}^3)$. Let

$$\mathbf{E} : \left(\text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}) \times X \right) \times \left(\text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R} \times \mathbb{R}^3) \right) \times [t_0, t_f] \rightarrow \mathbb{R}$$

be:

$$\mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) := \frac{1}{2} \int_D \rho(t, \cdot) \|\mathbf{u}(t, \cdot) - \tilde{\mathbf{u}}(t, \cdot)\|_{\mathbb{R}^3}^2 dx + \frac{1}{2} \|\rho(t, \cdot) - r(t, \cdot)\|_{L^2(D; \mathbb{R})}^2.$$

$\|\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)}$ with $f \in L^2(D; \mathbb{R}^3)$ is in fact a norm - it is obviously non-negative, $\|\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)} = 0$ iff $f \equiv_{L^2(D; \mathbb{R}^3)} 0$, and satisfies:

$$\|\rho^{1/2}(t, \cdot)(f + g)\|_{L^2(D; \mathbb{R}^3)} \leq \|\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)} + \|\rho^{1/2}(t, \cdot)g\|_{L^2(D; \mathbb{R}^3)}$$

$$\|C\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)} \leq |C| \|\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)},$$

so it is a norm over $L^2(D; \mathbb{R}^3)$, and indeed, using the constants $\tilde{\rho}, \bar{\rho}$ which we defined earlier in the proof of Theorem 3.9:

$$(\tilde{\rho})^{1/2} \|f\|_{L^2(D; \mathbb{R}^3)} \leq \|\rho^{1/2}(t, \cdot)f\|_{L^2(D; \mathbb{R}^3)} \leq (\bar{\rho})^{1/2} \|f\|_{L^2(D; \mathbb{R}^3)},$$

hence it is equivalent to the standard norm over $L^2(D; \mathbb{R}^3)$. Now, as in [30], we need to determine suitable remainder terms to construct estimates of $\mathbf{E}[\cdots](t)|_{t_0}^\tau, \tau \in [t_0, t_f]$. To that end, first assume that $(r, \mathbf{i})$ are also a bounded-energy weak solution to the Euler alignment system, i.e. they satisfy Definition 3.8 in addition to being a strong solution s.t.:

$$\partial_t r + \nabla_x \cdot \mathbf{i} = 0 \text{ a.e. in } [t_0, t_f] \times D.$$

We copy the energy inequality now expressed in $(\rho, \mathbf{u})$ (suppressing arguments):

$$\begin{aligned} \frac{1}{2} \int_D \rho \mathbf{u}^\top \mathbf{u} + \rho v_r[\rho] + \rho v_c[\rho] dx \Big|_{t_0}^\tau &\leq \dots \\ &\int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \pi_j[\rho \mathbf{u}] - \rho \mathbf{u}^\top \mathbf{u} \pi_\rho[\rho] + \rho \mathbf{u}^\top \nabla_x U dx dt + \dots \\ &\int_{t_0}^\tau \int_D \kappa_p \rho \mathbf{u}^\top \mathbf{u} \left(1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})\right) dx dt. \end{aligned}$$

We have:

$$\frac{1}{2} \int_D \rho v_r[\rho] dx \Big|_{t_0}^\tau = \int_{t_0}^\tau \int_D \partial_t \rho v_r[\rho] dx dt = - \int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \nabla_x v_r[\rho] dx dt,$$

and similarly for the term containing v_c , so the energy inequality becomes:

$$\begin{aligned} \frac{1}{2} \int_D \rho \mathbf{u}^\top \mathbf{u} \, dx \Big|_{t_0}^\tau &\leq \int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \pi_j[\rho \mathbf{u}] - \rho \mathbf{u}^\top \mathbf{u} \pi_\rho[\rho] + \rho \mathbf{u}^\top \nabla_x U \, dx \, dt + \dots \\ &\quad \int_{t_0}^\tau \int_D \kappa_p \rho \mathbf{u}^\top \mathbf{u} \left(1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})\right) \, dx \, dt + \dots \\ &\quad \int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \nabla_x v_r[\rho] \, dx \, dt + \int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \nabla_x v_c[\rho] \, dx \, dt. \end{aligned}$$

These computations were rather formal, but can be justified rigorously by testing on one hand the standard-mollified potentials $v_r^\epsilon := v_r[\rho] * \eta(\cdot; \epsilon)$ against the continuity equation and passing to the limit $\epsilon \rightarrow 0$, and on the other hand applying the fundamental theorem of calculus and radial symmetry of the convolution kernel.

To begin estimating $\mathbf{E}[\cdot \cdot \cdot](t)|_{t_0}^\tau$, $\tau \in [t_0, t_f]$, similarly to [31], first multiply the momentum equation by $\tilde{\mathbf{u}}$ and integrate:

$$\begin{aligned} \int_D \rho \mathbf{u}^\top \tilde{\mathbf{u}} \, dx \Big|_{t_0}^\tau &= \int_{t_0}^\tau \int_D \rho \mathbf{u}^\top \partial_t \tilde{\mathbf{u}} + \langle \rho \mathbf{u} \mathbf{u}^\top, \nabla_x \tilde{\mathbf{u}} \rangle_{\mathbb{R}^{3 \times 3}} \, dx \, dt - \dots \\ &\quad \int_{t_0}^\tau \int_D \rho \nabla_x^\top (v_r[\rho] + v_c[\rho] + U) \tilde{\mathbf{u}} \, dx \, dt + \dots \\ &\quad \int_{t_0}^\tau \int_D \rho \tilde{\mathbf{u}}^\top \pi_j[\rho \mathbf{u}] - \rho \mathbf{u}^\top \tilde{\mathbf{u}} \pi_\rho[\rho] \, dx \, dt + \dots \\ &\quad \int_{t_0}^\tau \int_D \kappa_p \rho \mathbf{u}^\top \tilde{\mathbf{u}} \left(1 - P(\|\mathbf{u}\|_{\mathbb{R}^3})\right) \, dx \, dt. \end{aligned}$$

Then, multiply the continuity equation by $\frac{1}{2} \tilde{\mathbf{u}}^\top \tilde{\mathbf{u}}$ and integrate:

$$\frac{1}{2} \int_D \rho \tilde{\mathbf{u}}^\top \tilde{\mathbf{u}} \, dx \Big|_{t_0}^\tau = \int_{t_0}^\tau \int_D \rho \tilde{\mathbf{u}}^\top \partial_t \tilde{\mathbf{u}} + \rho \mathbf{u}^\top (\nabla_x^\top \tilde{\mathbf{u}}) \tilde{\mathbf{u}} \, dx \, dt.$$

Again, these computations were a bit formal, but we can justify them as before using a mollification procedure on $\tilde{\mathbf{u}}$, and $\frac{1}{2} \tilde{\mathbf{u}}^\top \tilde{\mathbf{u}}$. The evolution of the $\rho \mathbf{u}^\top \mathbf{u}$ term is provided by the energy inequality. Then, combining these computations with the energy inequality, we obtain:

$$\mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \Big|_{t_0}^\tau \leq \int_{t_0}^\tau \mathbf{R}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \, dt$$

where $\mathbf{R} : \left(\text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}) \times X \right) \times \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R} \times \mathbb{R}^3) \times [t_0, t_f] \rightarrow \mathbb{R}$ is similarly to [30, 31]:

$$\begin{aligned} \mathbf{R}[(\rho, \mathbf{j}), (r, \mathbf{i})](\cdot) &:= \int_D \rho [\partial_t \tilde{\mathbf{u}} + (\nabla_x^\top \tilde{\mathbf{u}}) \mathbf{u}]^\top (\tilde{\mathbf{u}} - \mathbf{u}) \, dx + \dots \\ &\int_D \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x v_r [\rho] + \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x v_c [\rho] \, dx + \dots \\ &\int_D (\pi_{\mathbf{j}}^\top [\rho \mathbf{u}] - \mathbf{u}^\top \pi_\rho [\rho]) \rho (\mathbf{u} - \tilde{\mathbf{u}}) \, dx + \dots \\ &\int_D \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x U \, dx + \dots \\ &\int_D \kappa_p (1 - P(\|\mathbf{u}\|_{\mathbb{R}^3}) \rho \mathbf{u}^\top (\mathbf{u} - \tilde{\mathbf{u}})) \, dx. \end{aligned}$$

We use the Lipschitz solution to write:

$$\begin{aligned} \rho [\partial_t \tilde{\mathbf{u}} + (\nabla_x^\top \tilde{\mathbf{u}}) \mathbf{u}]^\top (\tilde{\mathbf{u}} - \mathbf{u}) &= \dots \\ \rho (\tilde{\mathbf{u}} - \mathbf{u})^\top (\nabla_x^\top \tilde{\mathbf{u}}) (\tilde{\mathbf{u}} - \mathbf{u}) &+ \rho (\partial_t \tilde{\mathbf{u}}^\top + \tilde{\mathbf{u}}^\top \nabla_x \tilde{\mathbf{u}}) (\tilde{\mathbf{u}} - \mathbf{u}). \end{aligned}$$

For the first term, since $\tilde{\mathbf{u}} \in \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}^3)$, it follows we can select some constant s.t. $\|\tilde{\mathbf{u}}\|_{\text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}^3)} < \mathbf{v}$.

$$\int_D \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x^\top \tilde{\mathbf{u}} (\mathbf{u} - \tilde{\mathbf{u}}) \, dx \leq 2\mathbf{v} \int_D \frac{1}{2} \rho \|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathbb{R}^3}^2 \, dx \leq C(\mathbf{v}) \mathbf{E}[\dots](t) \, dt.$$

For the second term, we insert:

$$\partial_t \tilde{\mathbf{u}} + (\nabla_x^\top \tilde{\mathbf{u}}) \tilde{\mathbf{u}} = \pi_{\mathbf{j}}[r \tilde{\mathbf{u}}] - \tilde{\mathbf{u}} \pi_\rho[r] - \nabla_x (v_r[r] + v_c[r] + U) + \kappa_p \tilde{\mathbf{u}} (1 - P(\|\tilde{\mathbf{u}}\|_{\mathbb{R}^3})).$$

The relative energy inequality becomes:

$$\mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \Big|_{t_0}^\tau \leq \int_{t_0}^\tau \tilde{\mathbf{R}}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) + C \mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \, dt$$

with:

$$\begin{aligned} \tilde{\mathbf{R}}[(\rho, \mathbf{j}), (r, \mathbf{i})](\cdot) &:= \int_D \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x v_r [\rho - r] + \rho (\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x v_c [\rho - r] \, dx + \dots \\ &\int_D (\pi_{\mathbf{j}}[\rho \mathbf{u} - r \tilde{\mathbf{u}}] - \mathbf{u} \pi_\rho [\rho] + \tilde{\mathbf{u}} \pi_\rho [r])^\top \rho (\mathbf{u} - \tilde{\mathbf{u}}) \, dx + \dots \\ &\int_D \kappa_p [(1 - P(\|\mathbf{u}\|_{\mathbb{R}^3}) \mathbf{u} - (1 - P(\|\tilde{\mathbf{u}}\|_{\mathbb{R}^3}) \tilde{\mathbf{u}})^\top \rho (\mathbf{u} - \tilde{\mathbf{u}})] \, dx. \end{aligned}$$

The function $\Phi_p(v) = v(1 - P(\|v\|_{\mathbb{R}^3}))$ is globally Lipschitz. Denote the Lipschitz constant by $L(\Phi_p)$. So, we obtain a bound on the final integral

using this Lipschitz constant:

$$\begin{aligned} & \int_D \kappa_p [(1 - P(\|\mathbf{u}\|_{\mathbb{R}^3}) \mathbf{u}^\top - (1 - P(\|\tilde{\mathbf{u}}\|_{\mathbb{R}^3}) \tilde{\mathbf{u}}^\top) \rho(\mathbf{u} - \tilde{\mathbf{u}}) \, dx \leq \dots \\ & 2\kappa_p L(\Phi_p) \int_D \frac{1}{2} \rho \|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathbb{R}^3}^2 dx = C(\kappa_p, P) \mathbf{E}[\dots](t) \, dt. \end{aligned}$$

Define $\mathfrak{B} : L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R} \times \mathbb{R}^3)) \rightarrow L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R}^3))$ as:

$$\mathfrak{B}[\rho, \mathbf{u}] := \pi_{\mathbf{j}}[\rho \mathbf{u}] - \mathbf{u} \pi_\rho[\rho].$$

By construction of $\pi_{\mathbf{j}}$ and via Fourier multipliers, for $(\rho, \mathbf{j})$ which also satisfy the definition of a bounded-energy weak solution in (3.2) - (3.4), and for $\mathbf{u} = \rho^{-1} \mathbf{j}$:

$$\pi_{\mathbf{j}}[\rho \mathbf{u}] \leq \frac{k_a}{\lambda_a} \|\rho \mathbf{u}\|_{L^2(t_0, t_f; L^2(D; \mathbb{R}^3))} = C(\bar{\rho}, k_a, \lambda_a) \|\mathbf{u}\|_{L^2(t_0, t_f; L^2(D; \mathbb{R}^3))}$$

and:

$$\mathbf{u} \pi_\rho[\rho] \leq C(\bar{\rho}, k_a, \lambda_a) \|\mathbf{u}\|_{L^2(t_0, t_f; L^2(D; \mathbb{R}^3))}.$$

Let Y be the set of functions which satisfy Definition 3.8. Since $\mathfrak{B}$ is bilinear over $L^2(t_0, t_f; L^2(D; \mathbb{R} \times \mathbb{R}^3))$ and bounded over $Y \subset \text{dom } \mathfrak{B} \subset L^2(t_0, t_f; L^2(D; \mathbb{R} \times \mathbb{R}^3))$, it is rather trivial to show that in any ball $B(0, \epsilon) \subset Y$, the functional $\mathfrak{B}|_Y$ (the restriction of $\mathfrak{B}$ to Y) is s.t.:

$$\begin{aligned} & \|\mathfrak{B}|_Y[\rho - r, \mathbf{u} - \tilde{\mathbf{u}}]\|_{L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R}^3))} \leq \dots \\ & L(\epsilon; \tilde{\rho}, \bar{\rho}, k_a, \lambda_a) \left(\|\rho - r\|_{L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R}))} + \|\mathbf{u} - \tilde{\mathbf{u}}\|_{L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R}^3))} \right). \end{aligned}$$

Now, for the alignment terms:

$$\begin{aligned} & \int_D (\pi_{\mathbf{j}}[\rho \mathbf{u} - r \tilde{\mathbf{u}}] - \mathbf{u} \pi_\rho[\rho] + \tilde{\mathbf{u}} \pi_\rho[r])^\top \rho(\mathbf{u} - \tilde{\mathbf{u}}) \, dx = \dots \\ & \int_D B[\rho - r, \mathbf{u} - \tilde{\mathbf{u}}]^\top \rho(\mathbf{u} - \tilde{\mathbf{u}}) \, dx. \end{aligned}$$

We can conclude using the previous result and the earlier equivalence of norms we showed earlier that:

$$\begin{aligned} & \int_D (\pi_{\mathbf{j}}[\rho \mathbf{u} - r \tilde{\mathbf{u}}] - \mathbf{u} \pi_\rho[\rho] + \tilde{\mathbf{u}} \pi_\rho[r])^\top \rho(\mathbf{u} - \tilde{\mathbf{u}}) \, dx \leq \dots \\ & L(\epsilon; \tilde{\rho}, \bar{\rho}, k_a, \lambda_a) \left(\|\rho - r\|_{L^2(t_0, t_f; L^2(D; \mathbb{R}))}^2 + \|\mathbf{u} - \tilde{\mathbf{u}}\|_{L^2(t_0, t_f; L^2(D; \mathbb{R}^3))}^2 \right) \leq \dots \\ & C(\dots) \mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \end{aligned}$$

where $\epsilon := \max\{\|(\rho, \mathbf{u})\|_{L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R} \times \mathbb{R}^3))}, \|(r, \tilde{\mathbf{u}})\|_{L^2(t_0, t_f; L^2(\text{cl } D; \mathbb{R} \times \mathbb{R}^3))}\}$. Finally, for the first term, via Young's convolution inequality and equivalence

of norms:

$$\begin{aligned} & \int_D \rho(\mathbf{u} - \tilde{\mathbf{u}})^\top \nabla_x v_r [\rho - r] \, dx \leq \dots \\ & \bar{\rho} \|\mathbf{u} - \tilde{\mathbf{u}}\|_{L^2(D; \mathbb{R}^3)} \|\nabla_x G(\cdot, k_r, \lambda_r)\|_{L^1(D; \mathbb{R}^3)} \|\rho - r\|_{L^2(D; \mathbb{R})} \leq \dots \\ & \frac{C(\bar{\rho}, k_r, \lambda_r)}{2} \left(\|\mathbf{u} - \tilde{\mathbf{u}}\|_{L^2(D; \mathbb{R}^3)}^2 + \|\rho - r\|_{L^2(D; \mathbb{R})}^2 \right) \leq \dots \\ & C(\bar{\rho}, \bar{\rho}, k_r, \lambda_r) \mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t). \end{aligned}$$

We perform a virtually identical procedure on the terms containing $\nabla_x v_c [\rho - r]$. We have successfully bounded all terms of the remainder by the relative energy. Finally, we have arrived at:

$$\mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \Big|_{t_0}^\tau \leq C(\dots) \int_{t_0}^\tau \mathbf{E}[(\rho, \mathbf{j}), (r, \mathbf{i})](t) \, dt,$$

with positive constant C independent of $\mathbf{E}$. The relative energy is obviously non-negative and initially zero. Since the inequality we have obtained holds for a.e. $t \in [t_0, t_f]$, by the Grönwall-Bellman inequality, we conclude that $\mathbf{E}[\dots](t) = 0$ a.e. on $[t_0, t_f]$. Since $\mathbf{E}$ is comprised of norms, the claims of Theorem 3.11 follow. This completes the proof. $\square$

4. GENERALIZATIONS AND COROLLARIES

In this section, we present two extensions to our model. First, we give a model of a swarm subject to an Euler alignment system which follows some controlled leader-agents whose position-velocity dynamics satisfy controlled ODEs. This gives a sparsely controlled system. We use our previous results to sketch a proof of existence of bounded-energy weak solutions and weak-strong uniqueness for this new model. Then, we present some results where the Yukawa potential of our original model has been replaced with Bessel potentials of fractional order $s > 2$, and similarly use our previous results and Bessel potential spaces to sketch proofs for existence of bounded-energy weak solutions and weak-strong uniqueness.

4.1. Sparse Control of the Density by Leaders. Consider ODEs for positions and velocities of leaders $(\xi_i, v_i) : [t_0, t_f] \rightarrow \mathbb{R}^3 \times \mathbb{R}^3$, $i = 1, 2, \dots, k$:

$$(4.1) \quad \begin{cases} \dot{\xi}_i(t) = v_i(t) \\ \dot{v}_i(t) = \mathbf{F}(\xi_i(t), v_i(t), f_i(t); \rho(t, \cdot)) \end{cases}$$

where $f_i \in \text{PWC}([t_0, t_f]; \mathbb{R}^3)$ is a control for the i -th leader, and $\rho \in C([t_0, t_f] \times D; \mathbb{R})$ a probability density in space at each time $t \in [t_0, t_f]$.

Suppose that $(v, \mathbf{F})$ leaves invariant the set $(\tilde{D} \subset D) \times B(0, s)$, $s > 0$ sufficiently large, with $\tilde{D}$ a nonempty, closed, connected, smooth subset, i.e.

$$v^\top \nu_\xi + \mathbf{F}^\top(\xi, v, f(t), \rho(t, \cdot)) \nu_v \leq 0 \text{ on } \partial(\tilde{D} \times B(0, s)) \, \forall t \in [t_0, t_f]$$

for f, ρ arbitrary. Then, if each $(\xi_i(t_0), v_i(t_0))^\top = (\xi_i^0, v_i^0)^\top \in \tilde{D} \times B(0, s)$, there is a unique classical solution to the ODEs bound to $\tilde{D} \times B(0, s)$ for

each i . Now, couple the given ODEs to the Euler alignment system through the potential U :

$$(4.2) \quad \begin{cases} \partial_t \rho + \nabla_x \cdot \mathbf{j} = 0 & \text{in } (t_0, t_f] \times D \\ \partial_t \mathbf{j} + \nabla_x \cdot [\rho^{-1} \mathbf{j} \mathbf{j}^\top] = \mathbf{S}[\rho, \mathbf{j}] & \text{in } (t_0, t_f] \times D \\ \Lambda_a \pi_\rho[\rho] = \mathbb{I}_D \rho & \text{in } \mathbb{R}^3 \\ \Lambda_a \pi_{\mathbf{j}}[\mathbf{j}] = \mathbb{I}_D \mathbf{j} & \text{in } \mathbb{R}^3 \\ \Lambda_c v_c[\rho] = \mathbb{I}_D \rho & \text{in } \mathbb{R}^3 \\ \Lambda_r v_r[\rho] = \mathbb{I}_D \rho & \text{in } \mathbb{R}^3 \\ \mathbf{S}[\rho, \mathbf{j}] := \rho \pi_{\mathbf{j}}[\mathbf{j}] - \mathbf{j} \pi_\rho[\rho] - \rho \nabla_x [\mathbf{1}_2^\top \mathbf{V}[\rho] + U(\cdot; \{\xi_i(\cdot)\}_{i=1}^k)] + \dots \\ \kappa_p \mathbf{j} (1 - P(\rho^{-1} \|\mathbf{j}\|_{\mathbb{R}^3})) \\ \mathbf{j}^\top \nu = 0 & \text{on } \partial D \times [t_0, t_f] \\ (\rho(t_0, \cdot), \mathbf{j}(t_0, \cdot)) = (\rho_0, \mathbf{j}_0) & \text{in } D. \end{cases}$$

We have the following result:

Theorem 4.1. *Consider the coupled ODE-PDE system (4.1, 4.2) and suppose that $U(\cdot; \{\xi_i\}_{i=1}^k)$ is additionally smooth and bounded w.r.t. $\{\xi_i\}_{i=1}^k$, and that:*

$$\sup_{x \in D, \{\xi_i\} \in D^k} \|\nabla_x U(x; \{\xi_i\}_{i=1}^k)\|_{\mathbb{R}^3} < \bar{U}_g,$$

$\bar{U}_g$ a positive constant. Then, assuming that the other hypotheses of Theorems 3.1, 3.9, 3.11 hold, their conclusions hold for this coupled ODE-PDE system.

Proof. (Sketch.) Let $\mathbf{x}_i[\mathbf{v}] = \xi_i(\cdot)$. $\mathbf{x}_i$ is constant w.r.t. $\mathbf{v}$, and is obviously Q -continuous, as is $U(\cdot; \{\mathbf{x}_i[\mathbf{v}]\}_{i=1}^k)$. Now, construct using the new structure of U the operator $\mathbf{M}[\cdot]$ using the Dirichlet problem for the given elliptic system (3.22). The operator $\mathbf{M}$ is still Q -continuous. So, the result of Theorem 3.1 follows. The proof of Theorem 3.9 need not be modified significantly, nor does the proof of Theorem 3.11. $\square$

Another result of the flexibility we have in these methods w.r.t. constructing momentum fields for densities of our choosing is the following result on reachability for the density:

Corollary 4.2. *Under the assumptions of the previous result, there are for every $r \in H^2(D; \mathbb{R}) \cap \text{Lip}(\text{cl } D; \mathbb{R})$ s.t. $r > 0$ in D controls $f_i \in \text{PWC}([t_0, t_f]; \mathbb{R}^3)$ s.t. $\rho(t_f, \cdot) = r$.*

Proof. (Sketch.) Pick some controls f_i . When we select the densities

$$\rho \in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}), \rho > 0, \rho(t_0, \cdot) = \rho_0$$

in Theorem 3.1, simply include the additional constraint $\rho(t_f, \cdot) = r$. Then, perform the Helmholtz decomposition on $\mathbf{j}$, and apply the previous results. $\square$

4.2. Bessel Potentials. We also have the following generalization to Bessel potentials:

Theorem 4.3. *Suppose Λ_α is of form:*

$$\Lambda_\alpha = \frac{1}{k_\alpha}(I - \nabla_x^2)^{\lambda_\alpha/2} \Leftrightarrow \mathcal{L}_\alpha = k_\alpha(I - \nabla_x^2)^{-\lambda_\alpha/2}$$

with $\lambda_\alpha \geq 2$. If the other assumptions of Theorems 3.1, 3.9, 3.11 hold, then their conclusions hold for this choice of operator.

Proof. (Sketch.) The sketch of the proof describes how to modify the arguments of Theorems 3.1, 3.9, and 3.11 to suit the new choice of non-local terms.

For $\lambda_\alpha = 2$, the non-local operator given above is the same type as the original choice, so we focus on $\lambda_\alpha > 2$. To start the arguments, as we've done before, pick arbitrary

$$\rho \in C^3([t_0, t_f]; H^2(D; \mathbb{R})) \cap \text{Lip}([t_0, t_f] \times \text{cl } D; \mathbb{R}), \rho > 0, \rho(t_0, \cdot) = \rho_0.$$

Define the Bessel potential spaces [50] for $s > 0, 1 < p < \infty$:

$$B^{s,p}(\mathbb{R}^3; \mathbb{R}) := \{f \in \mathcal{S}'(\mathbb{R}^3; \mathbb{R}) : \mathfrak{F}^{-1}[(1 + \xi^\top \xi)^{s/2} \mathfrak{F}[f]] \in L^p(\mathbb{R}^3; \mathbb{R})\}$$

with norm:

$$\|f\|_{B^{s,p}(\mathbb{R}^3; \mathbb{R})} := \|\mathfrak{F}^{-1}[(1 + \xi^\top \xi)^{s/2} \mathfrak{F}[f]]\|_{L^p(\mathbb{R}^3; \mathbb{R})}.$$

We have used $\mathfrak{F}[f]$ to be the Fourier transform of f , and $\mathfrak{F}^{-1}[f]$ to be the inverse Fourier transform of f . See [45] to review their properties. These spaces $B^{s,p}(\mathbb{R}^3; \mathbb{R})$ can be seen as interpolation spaces between the standard Sobolev spaces $W^{\text{ceil}(s),p}(\mathbb{R}^3; \mathbb{R})$ and $W^{\text{floor}(s),p}(\mathbb{R}^3; \mathbb{R})$ for non-integer s , and if s is an integer, $B^{s,p}(\mathbb{R}^3; \mathbb{R})$ coincides with a standard Sobolev space under an equivalent norm [50, 51]. Consider this definition to extend in the usual way to vector fields. For bounded, open, smooth sets $D \subset \mathbb{R}^3$, we define using restrictions [51]:

$$B^{s,p}(D; \mathbb{R}) := \{f \in \mathcal{S}'(D; \mathbb{R}) : \exists g \in B^{s,p}(\mathbb{R}^3; \mathbb{R}) \text{ s.t. } f = g|_D\},$$

with norm

$$\|f\|_{B^{s,p}(D; \mathbb{R})} := \inf_{g \in B^{s,p}(\mathbb{R}^3; \mathbb{R}), g|_D = f} \|g\|_{B^{s,p}(\mathbb{R}^3; \mathbb{R})}.$$

In particular, if $v \in L^\infty(D; \mathbb{R})$,

$$\mathcal{L}_\alpha \mathbb{I}_D v = k_\alpha G_{\lambda_\alpha}(\cdot) * \mathbb{I}_D v \in B^{\lambda_\alpha, p}(D; \mathbb{R}) \text{ for each } 1 < p < \infty,$$

where G_{λ_α} is the Bessel kernel [52]:

$$G_{\lambda_\alpha}(x) := \frac{1}{2^{\frac{3+\lambda_\alpha-2}{2}} \pi^{3/2} \Gamma(\frac{\lambda_\alpha}{2})} K_{\frac{3-\lambda_\alpha}{2}}(\|x\|_{\mathbb{R}^3}) \|x\|_{\mathbb{R}^3}^{\frac{\lambda_\alpha-3}{2}}$$

of order $\lambda_\alpha > 0$, and K_ν is the modified Bessel function of second kind. We have already selected ρ , so consider $\mathbf{j} \in X$, $\mathbf{j}(t_0, \cdot) = \mathbf{j}_0$, not necessarily a weak solution to (2.10) with the new interactions. $(\rho, \mathbf{j})^\top$ is regular

enough so that for $\lambda_a, \lambda_c, \lambda_r$ there are embeddings (after first embedding into $W^{\text{floor}(\lambda_a), p}(\dots)$):

$$\pi_{\mathbf{j}}[\mathbf{j}](t, \cdot) \in B^{\lambda_a, p}(D; \mathbb{R}^3) \subset C^{1, a}(D; \mathbb{R}^3) \text{ if } \mathbf{j} \in X, t \in (t_0, t_f)$$

and

$$\begin{aligned} \pi_\rho[\rho](t, \cdot), v_r[\rho](t, \cdot), v_c[\rho](t, \cdot) \in \dots \\ B^{\min\{\lambda_a, \lambda_c, \lambda_r\} + 2, 2}(D; \mathbb{R}^2) \subset C^{2, 1/2}(D; \mathbb{R}^2), t \in (t_0, t_f) \end{aligned}$$

for some $2 < p < \infty, 0 \leq a < 1$. Simply apply the arguments of Theorems 3.1, 3.9, 3.11 modified slightly for the new potentials and their conclusions follow. $\square$

5. LEARNING AGENT INTERACTIONS FROM DENSITY EVOLUTION IN 3D REGIONS WITH OBSTACLES

In this section, we make use of the Euler alignment system under study to discover the interaction (or coordination) dynamics (or laws) of a large swarm of agents interacting within a workspace cluttered with obstacles, from the observations of their density, i.e., their probability distribution (i.e. ensemble distribution) in three spatial dimensions, plus time.

We generate 3D particle trajectories using a version of the Boids model (Section 2, [15]). Such models have been used extensively to generate very realistic scenes and system trajectories and associated data. We bound the Boids to a cubical region of space with obstacles formed from cubes, and subject them to specular reflections. From these trajectories, we extract the histograms of the density and momentum profiles. We represent the initial mean-field distribution and momentum profile as a deep neural network for the initialization of the PDE describing the mean-field spatial probability (ensemble) distribution. The time-evolving spatial probability distribution estimates are then used as continuous observations of the density of the swarm. To discover the interaction (or coordination) laws, which we parametrize with a real vector, we pose a PDE-constrained optimization problem to minimize the squared Hellinger distance between the Boid histogram (i.e. the observed empirical ensemble distribution) and the hydrodynamic mean-field distribution. We solve this problem numerically and approximately using the Newton-conjugate gradient method. To the authors' knowledge, this is the first instance of a macroscopic multi-agent system identification problem for flocking being solved in three spatial dimensions in the presence of obstacles.

5.1. Problem Formulation. Similar to Section 2, we consider the sequence of particles $\{(x_i, v_i)\}_{i=1}^N(t) \subset (\text{cl } D \times \mathbb{R}^3), t \in [t_0, t_f]$, where $D \subset \mathbb{R}^3$ a bounded, open, connected, smooth region. Following the notation of Section 2, we assume that all agents are contained in a bounded polyhedral region $D \subset \mathbb{R}^3$. We assume that we have:

$$D_m := \{x \in \mathbb{R}^3 : \|x - \xi_m\|_\infty \leq R_m\}, \xi_1 = \mathbf{0}_{\mathbb{R}^3}$$

and $D_m \cap D_n = \emptyset, 2 \leq m, n \leq M, R_1 > R_m, 2 \leq m \leq M, \bigcup_{m=2}^M D_m \subset D_1$. Then, we define:

$$D := D_1 \setminus \bigcup_{m=2}^M D_m.$$

The interaction dynamics of the particles are given by the Boids model in (2.1), with the associated hydrodynamic system given by (2.10). In addition, we define some quantities. For two probability measures $P, Q \in \mathcal{P}(D)$ which are a.c. w.r.t. the Lebesgue measure on D , the squared Hellinger distance is:

$$H^2(P, Q) := \frac{1}{2} \int_D (\sqrt{p(x)} - \sqrt{q(s)})^2 dl(x),$$

where p, q are the densities (Radon-Nikodym derivatives) of P, Q w.r.t. the Lebesgue measure, respectively. Let $v_{max} := \max_{(t,i) \in [0,T] \times \{1, \dots, N\}} \|v^i(t)\|_\infty$, and define a sequence of regions: $\{\Omega_{ijk} \times \Upsilon_{lmn}\}_{i,j,k,l,m,n=1}^{\mathfrak{N}}$ which cover $[-R_1, R_1]^3 \times [-v_{max}, v_{max}]^3$ where Ω_{ijk} has volume $\Omega := (2R_1 \setminus \mathfrak{N})^3$, and Υ_{lmn} has volume $\Upsilon := (2v_{max} \setminus \mathfrak{N})^3$. Now, we define the position-velocity histogram over these cells as:

$$\mu^N(t, \Omega_{ijk}, \Upsilon_{lmn}) := \frac{1}{N} \sum_{\nu=1}^N \mathbb{I}_{(x,v) \in \Omega_{ijk} \times \Upsilon_{lmn}}(x^\nu(t), v^\nu(t)).$$

Via the histogram, we define a position-velocity density as:

$$m^N(t, x, v) := \dots$$

$$\frac{1}{\Omega \Upsilon} \sum_{i,j,k,l,m,n=1}^{\mathfrak{N}} \mathbb{I}_{(\xi,v) \in \Omega_{ijk} \times \Upsilon_{lmn}}(x, v) M^N(t, \Omega_{ijk}, \Upsilon_{lmn}),$$

associated to measure for $A \in \mathcal{B}([-R_1, R_1]^3 \times [-v_{max}, v_{max}]^3)$:

$$M^N(t, A) := \int_A m^N(t, x, v) dl(x, v).$$

The position histogram is:

$$q(t, x) := \int_{v \in \mathbb{R}^3} dM^N(t, x, v)$$

and the associated measure is:

$$Q(t, A) := \int_A q(t, x) dl(x),$$

with $A \in \mathcal{B}([-R_1, R_1]^3)$. For a solution to the PDE (2.10) at a particular time $t \in [0, T]$, define a measure (assuming $\rho(t, \cdot)$ is extended by 0 from D to D_1):

$$(5.1) \quad P(t, A) := \int_A \rho(t, x) dl(x)$$

for $A \in \mathcal{B}([-R_1, R_1]^3)$.

5.1.1. *Initial Conditions.* For the initial conditions, we assume that they are of the form:

$$(\rho_0, \mathbf{j}_0) = (\rho_0(\cdot; \theta), \mathbf{j}_0(\cdot; \theta))$$

where $\theta \in \mathbb{R}^d$ is a vector of neural network weights, i.e. the initial conditions are members of a family of neural network-parametrized functions. The neural network structure selected is that of [53] called the “Deep Galerkin Method” (DGM), which is similar to the LSTM structure [54]. The input is 3D, there are 2 DGM hidden layers with 50 DGM neurons each, and an output layer with 4 outputs, 1 for the density, and 3 for the momentum. This results in $d = 20350$ weights. The hidden layer neurons were *tanh* neurons, and the outputs were a *softplus* neuron for the density, and *tanh* neurons for the momentum. Let $P_0(\cdot; \theta)$ be the probability measure with $\rho_0(\cdot, \theta)$ as its density. The learning problem we solve for the initial conditions is:

$$(5.2) \quad \begin{aligned} \min_{\theta \in \mathbb{R}^d} L_1(\theta) &:= H^2(P_0(\cdot; \theta), Q(0, \cdot)) + \|\mathbf{j}_0(\cdot; \theta) - \hat{\mathbf{j}}_0\|_{\mathbb{R}^3}^2 \\ \text{s.t. } &\int_D dP_0(x; \theta) = 1, \rho_0(\cdot; \theta) \geq 0 \end{aligned}$$

where

$$\hat{\mathbf{j}}_0 := \int_{v \in \mathbb{R}^3} v \, dM^N(0, x, v).$$

This deep learning problem is solved in the typical way using the stochastic gradient method ADAM [55]. We scheduled the learning rate inversely w.r.t. the iteration number. The DGM network was structured so as to satisfy the constraints specified. The learned functions were used as the initial conditions for the PDEs in the system identification problem (5.3) posed below.

5.1.2. *System Identification.* In the model described by the system (2.10), there are $\hat{d} = 10$ parameters describing the agent interactions, which we denote by

$$\theta := (k_a, k_c, k_r, k_p, k_o, \lambda_a, \lambda_c, \lambda_r, \lambda_p, \lambda_o).$$

We pose the following PDE-constrained optimization problem:

$$(5.3) \quad \begin{aligned} \min_{\theta \in \mathbb{R}^d} L_2(\theta) &:= \frac{1}{T} \int_0^T H^2(P(t, \cdot; \theta), Q(t, \cdot)) \, dt \\ \text{s.t. } &(2.10), (5.1). \end{aligned}$$

We solve this problem numerically using a Newton-conjugate gradient method as we did in [6].

5.2. Numerical Methods.

5.2.1. *Numerical Integration.* All integrals in this work are approximated simply using Riemann integrals. Take the discretization of D to be given by $\{\Omega_{ijk}\}_{i,j,k=1}^{\mathfrak{N}}$, with centroids $\{s_{i,j,k}\}_{i,j,k=1}^N$. In (2.10), notice the sequence of elliptic problems. Via the fundamental solution, for example, the first problem is solved as:

$$\pi_\rho[\rho](t, \cdot) = \int_{\mathbb{R}^3} G(\cdot, s; k_a, \lambda_a) \rho(t, s) ds,$$

which, due to the structure of $G(\cdot; k_a, \lambda_a)$, is a singular convolution integral. With a slight abuse of notation, denote:

$$G(x, s; k_a, \lambda_a) = G(x - s; k_a, \lambda_a).$$

In discretization, the convolution integral becomes the Riemann (convolution) sum:

$$\pi_\rho[\rho](t, \cdot) = \sum_{i,j,k=1}^{\mathfrak{N}} \tilde{G}(\cdot - s_{i,j,k}; k_a, \lambda_a) \rho(t, s_{i,j,k}) \Omega$$

where (assuming $\lambda_a \neq 0$):

$$\begin{aligned} \tilde{G}(x - s; k_a, \lambda_a) = & \dots \\ & \begin{cases} 3k_a \frac{1 - \exp(-\lambda_a \frac{R_1}{2\mathfrak{N}})(\lambda_a \frac{R_1}{2\mathfrak{N}} + 1)}{\lambda_a^2 (\frac{R_1}{2\mathfrak{N}})^2} & \text{if } \|x - s\|_{\mathbb{R}^3} < \frac{R_1}{2\mathfrak{N}}, \\ G(x - s; k_a, \lambda_a) & \text{else .} \end{cases} \end{aligned}$$

so as to deal with the singularity when $x = s$. The first branch is constructed by averaging the value of $G(\cdot; k_a, \lambda_a)$ over a ball of radius $\frac{R_1}{2\mathfrak{N}}$ centered on the origin, i.e. where the singularity occurs in the kernel. To calculate the convolution sum efficiently, we take the very typical FFT-based approach.

5.2.2. *Numerical ODEs and Collision Event Handling.* To simulate the ODE system (2.1), we employ the velocity-Verlet [25] method, commonly used in particle dynamics. To deal with the collisions of agents with the boundary of the region D , we employ a typical approach taken in gas particle dynamics and molecular dynamics. The regions we deal with are cuboidal, so identifying if a particle has penetrated them in a timestep is rather simple. Between timesteps, particles are assumed to travel along a ray. To correct a boundary penetration, we trace this ray back to when it intersects the boundary of the region D . Again, as D is constructed from cubes, this is rather simple. The time when this occurs is also calculated, and the particle is marched in time up to the collision. Then, the particle's velocity is reflected about the surface normal, easily determined as the boundaries of the region D are entirely flat, and then the particle's position is marched up to the next timestep. Thus, the specular reflection condition specified is satisfied. Inter-agent collisions are not considered.

5.2.3. Numerical PDEs. In short, we employ a three-dimensional analogue of the second-order finite volume method we describe in our work in [8]. We do modify the fluxes in this method slightly. In this work, we use the Kurganov-Tadmor [34] flux with the minmod flux-limiter modified with the positivity-preserving flux-limiter of [56]. The positivity-preserving flux employed is the classical Rusanov-Local Lax-Friedrichs flux. In addition, we employ adaptive time-stepping so that the CFL number (the Courant-Friedrichs-Lewy condition number in CFD simulations, [57]) of the simulation is kept below .05. Time-marching is done via the strong stability-preserving second-order Runge-Kutta method [34].

There is also a small difference in the “dissipation” term $\mathbf{S}[\cdot]$ in this model. In our previous work, we only applied an alignment term. In this work, we deal with several more terms in the functional $\mathbf{S}[\cdot]$. The arrays which contain the discretizations of the terms of $\mathbf{S}[\cdot]$ are simply added together.

5.2.4. Choice of Parameters. For the workspace, we take 4 cubic obstacles, each of size $R_2 = \dots = R_5 = 1$. For the outer boundary, we let $R_1 = 5$. The centers of the obstacles were the columns of:

$$\begin{bmatrix} 2.5 & 2.5 & -2.5 & -2.5 \\ 2.5 & -2.5 & 2.5 & -2.5 \\ -4 & -4 & -4 & -4 \end{bmatrix}.$$

so each obstacle rests on the bottom of the arena. As stated earlier, timestepping is adaptive. The spatial mesh was taken to be $11 \times 11 \times 11$. We simulated 2000 Boids. Their initial conditions were sampled from a Gaussian, and re-sampled if they lay outside D .

5.3. Simulation Results. The training error in the optimization process is depicted in Fig. 1. In Fig. 2, the loss in our system identification problem is plotted against the iteration index. The Newton-conjugate gradient method brought quickly the PDE-constrained optimization problem we posed to a suboptimal solution. The Newton iterations terminated after 11 iterations. The final loss was $L_2(\theta_{11}) = 0.20008555429515784$.

The approximately optimal mean-field distribution does not become nearly as spread as the Boid distribution did. In the beginning, the squared Hellinger distance is very low, as we learned a representation of the initial conditions directly from Boid data. Then, as evolution starts, the squared Hellinger distance attains a peak, as in Fig. 3. This seems to be due to the spread of the Boid flock as opposed to the mean-field. The most concentrated regions of the Boid flock and the mean-field seem to coincide, as in Fig. 4. After this initial peak in divergence from each other, the Boid distribution and the mean-field become much more similar as time progresses. It is typical for the two flocks to appear as in Fig. 5, of course subject to some fluctuations in distance as shown in the plot of the Hellinger distance in Fig. 3. The times corresponding to Fig. 4 and Fig. 5 are highlighted with the blue and green dots, respectively, in Fig. 3. As mentioned in the

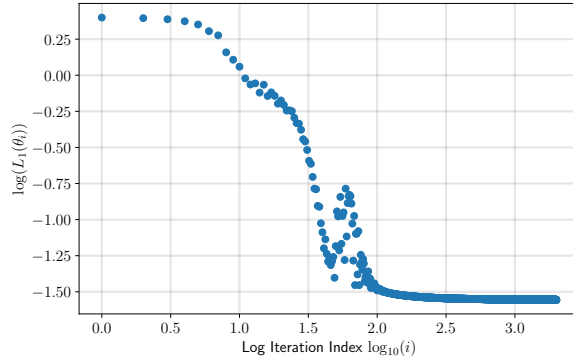


FIGURE 1. Training loss L_1 of the deep learning approximation problem (5.2) representing the initial conditions of the distribution of the agents simulated with the Boids model.

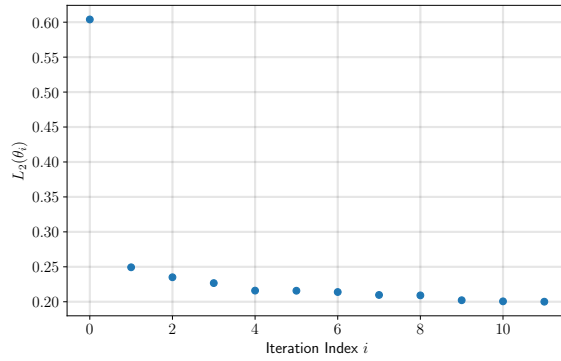


FIGURE 2. The loss functional in the PDE-constrained optimization problem (5.3). The initial descent steps of the Newton method are quite steep, in accordance to our observations in our previous work [7].

previous section, the spatial mesh was taken to be $(11 \times 11 \times 11)$, which was sufficient to recover the dynamics in sufficient detail - flocks with finer features or subflocks obviously require finer grids to capture.

6. CONCLUSION

In this work, we studied a (formal) hydrodynamic limit of a generalized Cucker-Smale model with

- i. a bounded domain (as opposed to free space or a torus);
- ii. singular alignment, repulsion, and cohesive potentials, in particular the Yukawa potential, a type of Bessel potential;
- iii. addition of a confinement potential;
- iv. incorporation of leaders and sparse control; and

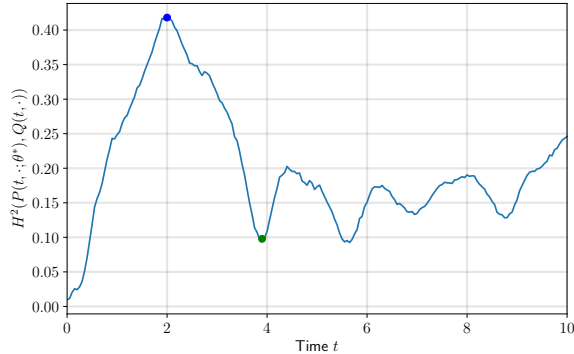


FIGURE 3. The squared Hellinger distance between the Boid distribution and the mean-field distribution at each simulation time. The mean-field distribution is interpolated linearly in time to match those of the Boid simulation as it is timestepped according to its own stability criteria. We have highlighted two points. In blue, we highlight the time (time instant $t \simeq 2$) at which the boid and mean-field distributions are the most different. In green, we highlight a time (time instant $t \simeq 4$) when they are more similar. We plot these distributions in Fig. 4 and Fig. 5, respectively.

v. a generalization to Bessel potentials of arbitrary order.

More specifically, unique Lipschitz densities were admitted by the Euler alignment system, and solutions with Lipschitz momenta are unique in the class of bounded-energy weak solutions. Assuming Lipschitz solutions exist for arbitrarily long times, we may use these conclusions *a priori* in system identification problems and sparse (optimal) control problems for biological and artificial swarms, respectively. In particular, we made use of the proposed analysis of the Euler alignment system under study to design and solve a PDE-constrained optimization problem to identify the interaction dynamics of a simulated swarm of agents from observations of their density (swarm ensemble distribution) evolution in 3D regions with obstacles.

As mentioned the identification of coordination laws of swarms from partially observed data (microscopic or macroscopic) is a motivating key problem studied in recent previous works by the authors [7, 8, 24]. As also mentioned the distributed robust control of swarms is another motivating key problem studied in previous works by the authors and others [11, 12, 13, 14, 6, 9, 10]. The non-uniqueness of solutions to the inverse problem (identification of coordination laws) established in our earlier works leads to the problem of identifying the simplest possible coordination laws from microscopic and macroscopic data. The solution to this problem has important implications on the efficiency, response speed and accuracy, of the resulting solutions to both motivating problems. One interesting problem

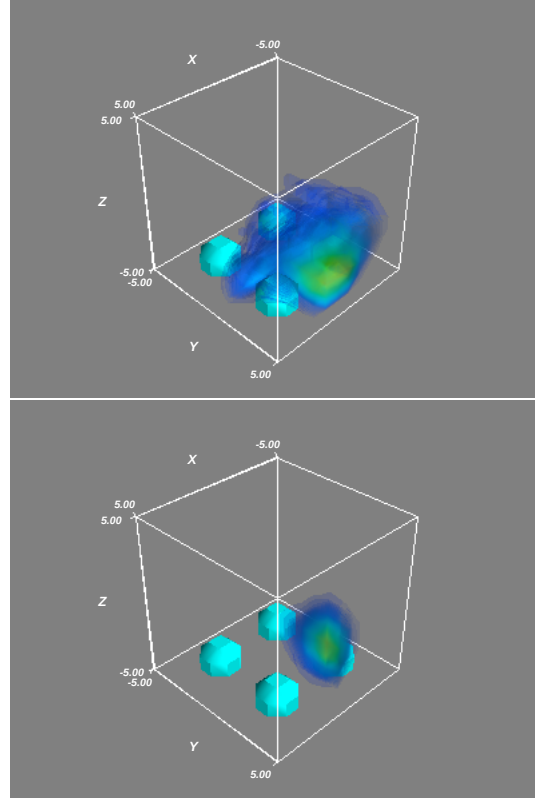


FIGURE 4. (top) The Boid distribution at the time (time instant $t \simeq 2$) of maximal Hellinger distance from the mean-field distribution. (bottom) The mean-field distribution at the same time. This is labelled with a blue dot in Fig. 3.

for example is to understand the simplest distributed communication scheme in such multi-agent systems (including both direct and indirect (via sensing or motion) communication between agents). Closely related problems are the robust distributed control of such multi-agent systems, the reliability of the resulting coordination schemes, and the self organization of such networked systems for efficient autonomous and semi-autonomous operation. In the present paper we provided the analysis of the dynamics of models comprised of linked PDEs and ODEs for such swarms, using several potentials with explicit functional forms. Extensions of these results to general forms of potentials and the analysis of the robustness and reliability of the resulting dynamics when using such general potentials is a key problem that needs further investigation. These problems are under current and future investigation by the authors.

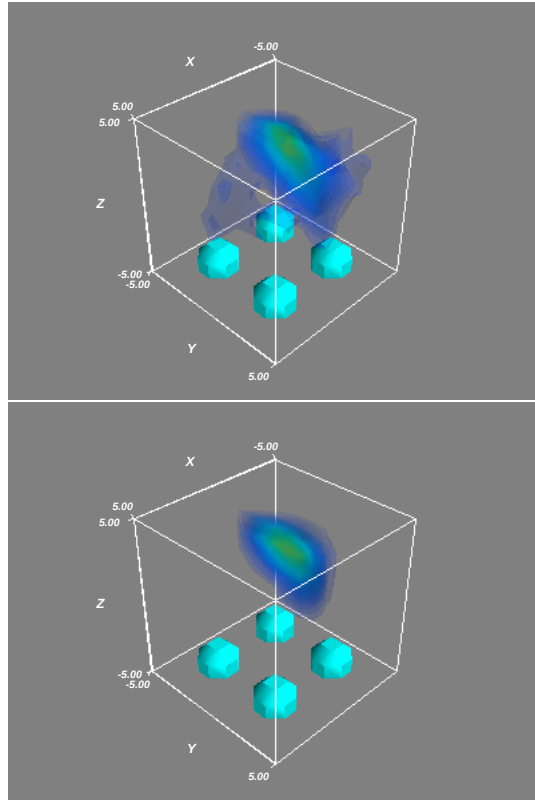


FIGURE 5. (top) The Boid distribution at a time (time instant $t \simeq 4$) of lower Hellinger distance from the mean-field distribution. (bottom) The mean-field distribution at the same time. This is labelled with a green dot in Fig. 3.

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